

It from bit: a concrete attempt

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Abstract

We introduce a fully specified deterministic cellular automaton as a concrete test of Wheeler’s “it from bit” idea. Space is a three-dimensional torus with periodic boundaries, plus a non-spatial internal dimension split into two sectors. Each cell stores a fixed-length binary word that evolves through local rules built from classical logic and elementary arithmetic. Wave propagation is limited to a finite speed by updating cells in discrete layers. The model produces structures that resemble familiar physics. Waves obey a discrete wave equation with a tunable dispersion relation. Stable, self-reinforcing patterns behave like particles: they move, collide, exchange momentum, and can be created or destroyed in pairs, while conserving global counts. Interactions among these patterns show signs of attraction, repulsion, and orbit-like bound states. Probabilistic-looking behavior arises from deterministic rules because the finite state space can be coarse-grained in many equivalent ways. A built-in temporal direction comes from irreversible update steps and memory buffers. Charge-like quantum numbers appear as conservation laws tied to internal-sector symmetries; particle-antiparticle pairing arises from complementation rather than from any built-in asymmetry. We carefully label each claim as a postulate defining the model, a measured result of simulations, or a conjecture still to be proved. We do not claim to recover known physics; we report a complete discrete toy universe and the ideas proposed to close the remaining gaps. The scale of the lattice is not derived: the one place where it is matched to observation is an order-of-magnitude calibration, conditional on identifications stated as such.

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“It always bothers me that, according to the laws as we understand them today, it takes a computing machine an infinite number of logical operations to figure out what goes on in no matter how tiny a region of space, and no matter how tiny a region of time. Why should it take an infinite amount of logic to figure out what one tiny piece of space-time is going to do? So I have often made the hypothesis that ultimately physics will not require a mathematical statement, that in the end the machinery will be revealed, and the laws will turn out to be simple, like the checkerboard with all its apparent complexities.”

Richard Feynman

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1 Introduction

John Archibald Wheeler’s ‘it from bit’ thesis proposes that the physical world may emerge from discrete information, with bits as the bedrock of reality [1]. Cellular automata (CA) — discrete models of space, time, and local rules — provide a natural setting for exploring this idea, since simple rules can produce complex, physics-like behavior [2, 3, 4].

This work presents a deterministic cellular-automaton universe built from classical logic, elementary arithmetic, and a minimal 3-torus topology augmented by a non-spatial internal dimension. Each cell carries a fixed-size binary string whose ontological state evolves by local rules, producing spherical wavefronts, harmonic modes, and symmetry-breaking patterns. Wave propagation is confined to a spherical cavity inside the torus, suppressing corner artifacts while preserving periodic boundary conditions. We examine how these emergent structures reproduce features reminiscent of particles, interactions, and spacetime organization. The model is not offered as an interpretation of quantum mechanics, but as a primitive descriptive layer.

Claims and status. To keep claims commensurate with evidence, every substantive statement below is graded as one of: (P) a postulate that defines the rule; (M) a measured result of the reference implementation on a small lattice; or (C) a candidate mechanism or conjecture that the current implementation does not yet establish. The model-definition sections contain the postulates; the particle taxonomy, the quantum-formalism mapping, and the prospects are labeled conjectural in the full manuscript. The paper does not claim that this automaton reproduces physics: it reports a fully specified discrete toy universe, the quantitative tests it fails or passes, and the candidate mechanisms proposed to close the gaps. The only place where the dimensional parameters of the fabric are matched to observation is Appendix A: it is a calibration conditional on identifications stated there, its numbers are order-of-magnitude values, and the single observational test it applies rejects one of the two readings rather than validating the other.

One interpretive choice governs the entire construction and should be stated from the outset. The constant velocity at which spherical wavefronts expand — one lattice cell per tick — is *not* the speed of light. It is the maximum speed of information transmission, $v_{\max}$, a fundamental constant of the fabric. The speed of light c is emergent: it is the value that an observer assigns to $v_{\max}$ under a round-trip measurement convention. Thus c is a property of the measuring structure, not of the transition rule. Conversely, some operations of the model — encounter and the other globally coordinated updates — are superluminal in coordinate terms, yet they carry no capacity to transmit information and cannot be manipulated by any observer. In the idealised local rule, causality is thereby protected by construction: every influence that can be emitted, read, or steered propagates at or below $v_{\max}$. For the host-side scheduler of the present implementation this remains to be established; the caveat is restated where non-local coordination is introduced (Sect. 6).

Our construction extends foundational work in digital physics: Zuse’s discrete universe [3], Fredkin’s reversible computation [4], and Wolfram’s complex behavior from simple rules [2]. The proposal is a layered 3-torus with explicit interaction rules; whether those rules reproduce quantum-like or relativistic effects is the question this paper tests, not an assumption.

The paper is organized as follows. Section 2 reviews previous work. Section 3 defines the universal fabric, and Section 4 builds the initial state. The phase step and the emergent wave fields are described in Section 5; the light frame in Section 6. Section 7 details bubble interactions; Section 8 addresses the dynamic organization of W -islands. Section 9 closes with a synthetic interpretation.

Source and supporting material

The simulator of this document is the program `it_from_bit`, published at https://github.com/automaton3d/it_from_bit: the CPU build with the GUI, the rule set under `src/model/` and the framework under `src/`. The study archive that the text points to — the experiment logs and derivations under `experiments/`, the flux harness under `quantization/`, the retired campaigns under `attic/`, the helical-walker visualization `spiral/` and the charge-algebra studies `combine.c` and `virada.c` — is maintained in the companion repositories of the same program and is *not* part of this one. The file paths quoted in the text therefore locate material in that archive, not files distributed here; where a claim rests on such material, the argument in the text is meant to stand on its own.

Glossary of proprietary terms

For readability, the following model-specific terms are collected here with one-line definitions; the section in brackets is where each is first defined or used.

Term	Meaning	Sect.
Orbis / Umbra	the two sectors of the internal dimension, indexed by the weak bit w_1 (Sect. 3.1); each cell’s charge word carries one sector bit	3.1
bubble	a spherical wavefront emitted by one source center in one layer w ; the elementary extended object of the model	6.1
era	one complete wavefront expansion-and-contraction cycle; distinct from a light frame and from a housekeeping tick k	6
chief (K)	the leading source of a W -island; delegates (D) are attached to it, singletons (S) are free sources, and a pair (P) is a charge-complementary bonded pair of sources	6.1
drive pair	an affinity-bound photon-type pair carrying $\mathbf{m} \neq \mathbf{0}$; its $P \times K$ and $P \times D$ contacts sustain island motion; it cannot join a cluster	6.1.2
cluster	a group of superposed equal-status pairs with the gluon/photon charge geometry $R2$	6.1
homing	the <i>homB</i> flag stage of the light frame that drives the relocation vector c toward a target	6.7
$s2B$ sieve	the electroweak gate $((u \cdot t) \bmod S) < u$ that admits a fraction of the overlapping active cells to the interaction branch	5
Hofer effect	the conjectured radial alignment of the spins of a localized particle (Sect. 4)	4

2 Related work

The hypothesis that the physical universe may be fundamentally discrete and computational has a long history in theoretical physics. Our construction builds upon several foundational programs in digital physics, extending them with a specific layered topology and interaction ruleset.

Early digital physics and reversible computation

The notion of a computing universe traces back to Zuse [3], who proposed in *Rechnender Raum* that space and time are discrete and that physical processes arise from local, rule-based state transitions on a grid of finite-state machines. Zuse introduced “digital particles” as localized, self-reproducing lattice patterns and explored how continuum physics might emerge from discrete dynamics. This vision was formalized by Fredkin [4] through the *Finite Nature Hypothesis*, which posits that the universe is a cellular automaton (CA) whose evolution strictly preserves information.

Because standard CA updates are often dissipative, Fredkin and Margolus [5] developed reversible computation frameworks. Margolus introduced block cellular automata (the Margolus neighborhood), demonstrating that permutations on shifting blocks can conserve mass, energy, and momentum directly from bit-level dynamics. These lattice-gas models proved that discrete, local, invertible rules can reproduce continuum equations at coarse-grained scales, establishing that CA-based frameworks are capable of capturing key symmetries of physical law.

Modern interpretations and hypergraph models

More recently, the discrete hypothesis has been invoked to address foundational issues in quantum mechanics and spacetime. ’t Hooft [6] proposed a deterministic CA interpretation of quantum mechanics. In that interpretation, quantum states are coarse-grained equivalence classes of deterministic ontological CA states, and the Hilbert-space formalism is reconstructed from classical CA dynamics via information loss at the smallest scale.

Concurrently, Wolfram [2, 7] has championed the computational universe hypothesis, arguing that simple CA rules (such as Rule 110) can generate the complexity required for universal computation. His recent Wolfram Physics Project generalizes CA from grid cells to hypergraph nodes updated by rewrite rules, aiming to derive spacetime and relativistic invariance from pure graph rewriting.

Lattice-gas and lattice-Boltzmann automata

A distinct tradition uses reversible block cellular automata as lattice gases that reproduce continuum hydrodynamics from purely local integer rules. Hardy, Pomeau and Pazzis introduced the HPP lattice gas [8]; Frisch, Hasslacher and Pomeau showed that a hexagonal lattice removes the spurious anisotropy of the square HPP model and yields the Navier–Stokes equation in the macroscopic limit [9]. The FHP model is the paradigmatic example of a deterministic, local, integer rule whose coarse-grained dynamics are those of a continuous field theory. The present work shares that philosophy—a strictly local, integer-only rule with emergent continuous behavior—but differs in the target and in the ontological status of the continuum. A lattice gas is a discretization of a pre-existing field: the hydrodynamic limit is a theorem, and the discrete objects are computational devices, not the particles themselves. Here, by contrast, particle-like objects (pairs of bubbles riding spherical wavefronts) are meant to emerge directly from the microscopic rule, without an intermediate continuum. The price of that ambition is that our continuum limit remains a labeled conjecture rather than a proven limit. The gain claimed is precisely the “strict deterministic ontology without coarse-graining” of Sect. 3.3. Nothing in the rule refers to a fluid, a field, or a metric, and the emergent c is a reading of the fabric by an observer rather than a parameter of a coarse-grained equation.

Positioning of the present work

While these foundational works establish the plausibility of a computational substrate, they often leave the explicit mapping between bit-level rules and emergent particle phenomenology as an open program. Our work differs by proposing a concrete, strictly local, integer-arithmetic rule set on a 3-torus augmented by an internal dimension W . Unlike ’t Hooft’s equivalence classes or Wolfram’s hypergraphs, we maintain a strict deterministic ontology without coarse-graining, attempting to derive spherical wavefronts, polarization, and interaction gates (such as the electroweak sieve) directly from the microscopic transition rules.

3 The universal fabric

Let L be a large integer. The reference implementation restricts the simulated lattices to odd edges, so that the seed has a single central cell (Sect. 4.1); the physical calibration of App. A takes L to be a multiple of 64, so that the internal ledger closes with integer copies. The fundamental fabric of the model is a discrete lattice of cells arranged in a 3-torus ($\mathbb{T}^3$) topology. This finite, closed space is periodic: each boundary wraps around to its opposite edge, ensuring global continuity without physical boundaries.

Beyond the three spatial dimensions, the framework includes a compact internal dimension of W discrete positions (referred to as *layers* or *slots*). Positions in W do not represent additional spatial directions; rather, they serve as internal addresses that group bubbles by shared charge bits (Sect. 4.1). The internal organization in W is essential for the continuous dissociation and reconstruction of collective structures. Stable three-dimensional particles emerge as coherent islands whose constituent bubbles remain correlated in physical space. During scattering or annihilation events, these islands may dissolve into individual bubbles, which subsequently reorganize through new pairing processes, allowing new collective structures to emerge while conserving the total bubble population.

The size of the internal dimension is defined as

$$W = 3L^2. \tag{1}$$

Its quadratic scaling naturally matches the characteristic cross-sectional scaling of the discrete three-torus and provides sufficient internal degrees of freedom for the continual pairing, dissociation, and reorganization of bubbles. A dynamical motivation for this scaling, together with the internal population $W = 3L^2$ and the declared partition of its addresses into N_I charge families, is given in Section 8.

3.1 The cell structure

Each cell carries the same fixed-size bit string, formatted as shown in Table 1. Evolution progresses in uniform steps, with the input and output ports of each cell conceptually linked through a Finite State Machine (FSM). The cell state is partitioned into physical properties, wavefront shaping variables, and operational variables.

Physical properties. The physical role of the cell state is summarized in Table 1.

- **Charge** (*ch*): A six-bit word encoding discrete properties that govern interactions and force emergence (detailed in Sect. 3.4).
- **pBit** (*pB*): Marks the local in-phase sign of the emergent polarization pair ($\text{pol}_u, \text{pol}_v$) reconstructed from the broadcasted arrival stamp $b(x)$. It is true when the in-phase component pol_u is positive ($pB = (\text{pol}_u > 0)$), triggering the electric interaction channel.
- **sBit** (*sB*): Signals the emergent transverse polarization of the cell. It is true when the reconstructed transverse component pol_v is positive ($sB = (\text{pol}_v > 0)$).
- **Affinity** (*a*): An integer variable that groups layers into particles, facilitating the aggregation of smaller units into complex entities. It is the substrate for the emergent phenomenon of entanglement and characterizes the track left by particles in self-interference.

Wavefront shaping. These variables form the mechanism for nearly perfect spherical wave propagation under discrete constraints.

- **Dynamic squared radius** (r^2): The squared Euclidean distance from the current wave source, updated locally by an incremental rule through the 6-neighborhood. It is INF before the wavefront reaches the cell.
- **Dynamic radius** (r): The integer radius propagated alongside r^2 and corrected by local square-boundary tests; it is used in the wave equation, shell forcing, and boundary absorption.
- **Wavefront active** (*active*): A boolean flag true when the cell currently lies on the pulsating spherical wavefront. It is computed by comparing the cell's propagated integer radius r with the local frame radius $\text{effective_t}(t)$, which advances one cell per frame at the maximum information speed $v_{\max}$.
- **Wave phase** (f): The triangular breathing phase of the wavefront, $f = \text{effective_t}(t)$. It rises one cell per frame from 0 to $L/2$ on the expansion branch of an era and falls back to 0 on its contraction branch (Sect. 6).
- **Wavefront tick** (t): A local physical clock providing a step counter for the maximum speed of information transmission $v_{\max}$.

Operational variables. The model performs operations—translation, encounter, and wavefront propagation—that must be coordinated across the lattice. Such coordination is supported by the distinction between two clocks. The housekeeping tick k is incremented synchronously in every cell and acts as an atemporal, global phase clock for the FSM. The wavefront tick t , on the other hand, is a local physical clock: it is reset at each source and may differ across layers and regions. The interplay between the common k and the local t allows the framework to schedule essentially non-local processes (encounter and coordinated updates) without endowing them with any capacity to transmit information: such processes are superluminal in coordinate speed and are intended to be strictly non-signaling, protecting causality. For the host-side scheduler used by the current implementation, this non-signaling property is a design goal rather than an established fact (see the full manuscript).

- **Translation offset** (c): An integer vector determining the reissue address, guiding the position where an element is reallocated within the system. The displacements follow the general rule $d_i = (t_i - c_i + L) \bmod L$, where $c_i = \lfloor (L - 1)/2 \rfloor$ is the central cell of the axis (Sect. 3).
- **Housekeeping tick** (k): An atemporal clocking mechanism used for maintaining internal processes.
- **Sieve test result** (*s2B*): A Boolean bit seeded at each wave step from the local wave displacement u and the step counter t . For $u > 0$ on an active cell, the deterministic test $((u \cdot t) \bmod \text{SHELL_TARGET}) < u$ produces a trigger, where $\text{SHELL_TARGET} = 16384$. The stored value is gated by the active flag, $s2B = \text{active} \wedge \text{trigger}$. Electroweak interaction occurs only when this bit is true.
- **Collapse flag** (*kB*): Indicates the occurrence of collapse in the interaction, warning the components of a particle.
- **cluster flag** (*bB*): Indicates whether the pair can participate in a cluster (a group of superposed pairs with the same affinity, same *bB* bits, and the same active *pB*).

Table 1: Cell state structure. Types: N (integer), DN (double integer), V (3D integer vector), B (boolean), B6 (six-bit word). The charge word ch is ordered $w_1 w_0 q c_2 c_1 c_0$ (most significant bit first), matching the implementation of Ref. [10]. Bit values: $q \in \{+, -\}$; $w_1 \in \{O \text{ (Orbis)}, U \text{ (Umbra)}\}$; $w_0 \in \{R \text{ (Right)}, L \text{ (Left)}\}$; $c_2 c_1 c_0 \in \{N, R, G, \bar{B}, B, \bar{G}, \bar{R}, \bar{N}\}$.

Name	Type	Symbol	Observations
Physical properties			
charge	B6	ch	bits $w_1, w_0, q, c_2, c_1, c_0$
pBit	B	pB	local in-phase sign, $pB = (\text{pol}_u > 0)$
sBit	B	sB	emergent transverse polarization, $sB = (\text{pol}_v > 0)$
affinity	N	a	groups bubbles into particles
position	V	$\mathbf{x}$	spatial coordinates
Wavefront			
dynamic squared radius	DN	r^2	squared distance from the wave source
dynamic radius	DN	r	integer Euclidean radius from the source
wavefront active	B	$active$	true when $r = \text{effective.t}(t)$
wavefront tick	DN	t	clocking of the maximum information speed
wave phase	DN	f	triangular breathing phase $f = \text{effective.t}(t)$
broadcast stamp	DN	$b(x)$	arrival tick of the elected-momentum news
polarization u	N	pol_u	in-phase reconstructed polarization component
polarization v	N	pol_v	transverse reconstructed polarization component
Operational variables			
translation offset	V	$\mathbf{c}$	determines reissue address
momentum vector	V	$\mathbf{m}$	integer momentum, typically $ \mathbf{m} = L/2$; immutable under inertia
translation impulse	V	$reloc$	consumable signed displacement; filled by encounter, applied by <code>appl</code>
housekeeping tick	DN	k	timeless clocking
sieve test result	B	$s2B$	seeded from u and t , gated by $active$
sB homing	B	$homB$	flag to find the sB bit
contraction flag	B	cB	marks a bubble in the contraction phase
collapse flag	B	kB	hints collapse
cluster flag	B	bB	marks pairs eligible for cluster formation
Source model			
source kind	N	kind	K (chief), S (singleton), D (delegate), P (pair)
pair index	DN	pair_idx	partner layer of a P source
pair count	N	pair_count	superposed pairs of a P source

3.2 The host lattices

The spatial fabric is implemented using three coexisting 3D lattices of L^3 cells each, replicated across the W layers (totaling $3L^3W$ cells). All cells within a layer evolve synchronously, exchanging information with their seven neighbors (six spatial, one along W). This exchange occurs in alternating passes, recovering time homogeneity every two clock ticks and configuring the system as a cellular automaton. The three lattices are referred to as *main*, *draft*, and *partner*. The main lattice is the current state used in interactions; the draft lattice holds the modified data for the current tick; and the partner lattice is a snapshot used for comparisons across different W addresses during the encounter stage. A cell can modify its own draft cell, but not its main cell directly.

3.3 Purity constraints

The rules of this universe are deliberately constrained so that the model remains a *pure* cellular automaton. These constraints shape the physics that the automaton can produce and are recorded here as design criteria.

Locality. A cell's next state is computed only from its own current state and from the states of a fixed, finite neighborhood of adjacent cells. No cell reads global counters, hidden pointers, queues, frontier lists, or pre-computed tables that depend on the whole lattice.

Synchronicity and parallelizability. Every cell updates simultaneously from the current lattice into a second, identical lattice (`grid` and `grid.next`). The two buffers are the only shared memory; after the tick, the roles are swapped. This makes the rule directly parallelizable.

Integer-only runtime. Inside the transition rule, all calculations use only addition, subtraction, bit shifts, and comparisons. Multiplication, division, floating-point arithmetic, square roots, trigonometric functions, and any other continuous operation of *dynamic* operands are forbidden in the runtime loop. Fixed topological constants (e.g., $2R$, $L/2$, N_I) are evaluated once at initialization and stored in read-only integer globals. Multiplication by such a constant is a fixed-strength operation, not the data-dependent multiplication the constraint excludes.

No tables. No pre-computed lookup tables encode the dynamics. Distances, phases, and interaction weights must propagate through local rules from the initial seed.

Finite-state and deterministic. Each cell contains a fixed-size bit string. The transition rule is a finite function of this string and its neighborhood, with no Turing-complete constructs, no unbounded iteration, and no randomness.

Occam's razor. The cell state is kept minimal. Auxiliary global variables, linked lists, or index arrays that point into the lattice are rejected because they would make the automaton's state depend on the order of evaluation rather than on the cell contents alone.

Simple initialization. The initial state is set by fiat only at $t = 0$: a small seed is planted in an otherwise blank lattice. All subsequent structure emerges from the local rule.

These are target constraints, not properties already verified for the executable. In particular, the current inertial transport reference uses source snapshots, contact lists, a frame counter and periodic layer translations (Sect. 6.5); polarization preparation also uses host arithmetic. A local finite-state realization of this coordination remains open. The theoretical topology is parameterized by L and $W = 3L^2$, while the prepared-island experiments may allocate a reduced number of layers.

3.4 Charges

Charges are constant-valued bits (*ch* in Table 1) that govern particle interactions. Inspired by the Standard Model, the six-bit charge word is partitioned into electric, weak, and color sectors. All cells in a given W -layer share the same charge value.

The electric charge q is associated with attraction and repulsion, while the weak charge, composed of two bits w_1 and w_0 (chirality), is associated with congruence. The sector with $w_1 = 0$ is called *Orbis*, while the sector with $w_1 = 1$ is called *Umbra*. The three color charges c_2, c_1, c_0 enforce the strong force structure. Color codes are $N : 000$, $R : 001$, $G : 010$, $\bar{B} : 011$, $B : 100$, $\bar{G} : 101$, $\bar{R} : 110$, $\bar{N} : 111$ for

neutral, red, green, antiblue, blue, antigreen, antired, and anti-neutral, respectively. Figure 1 draws the field layout and the full code table.

Let the signature be $sig = c_2 + c_1 + c_0$. A bubble is matter M if $sig < 2$, or antimatter $\bar{M}$ otherwise. It is neutral N if $sig = 0$, or anti-neutral $\bar{N}$ if $sig = 3$. The pBit pB marks the positive in-phase half-cycle of the polarization pair, while the sBit sB records the transverse sign derived from the reconstructed (pol_u, pol_v) pair. Both emerge from the cellular dynamics rather than being precomputed path patterns.

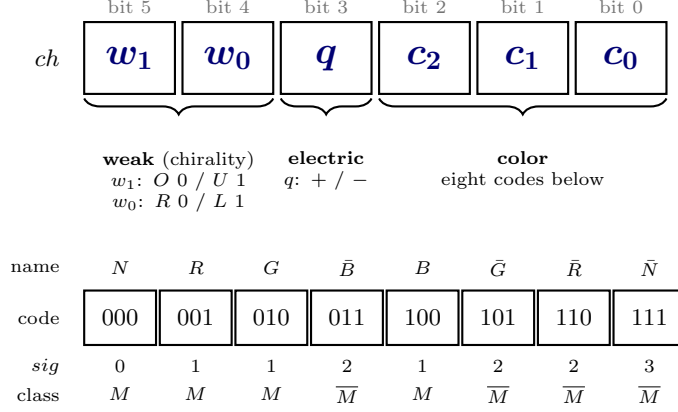


Figure 1: The charge word. The six bits of ch , most significant first (Table 1), and the three sectors they encode: the two weak (chirality) bits, of which w_1 names the sector (Orbis 0; Umbra 1) and w_0 the chirality (Right 0; Left 1); the electric bit q ; and the three color bits $c_2c_1c_0$. The strip below lists the eight color codes with their signature $sig = c_2 + c_1 + c_0$ and the class it fixes: matter M for $sig < 2$, antimatter $\bar{M}$ for $sig \geq 2$, with $sig = 0$ the neutral code N and $sig = 3$ the anti-neutral code $\bar{N}$. Because sig counts the color bits that are set, complementing the color word maps $M \leftrightarrow \bar{M}$: the word and its complement sit at opposite corners of the strip. Every cell of one W -layer carries the same word (Sect. 3.4), and the seed’s family map (Sect. 4) derives all four fields from the family index, so only eight of the sixty-four words are realized.

4 Initial state

The initial state is the only information set by fiat. It fixes the static, non-dynamical configuration of every cell at $t = 0$ and is copied to the draft lattice as well. All dynamic quantities—the wavefront radius, the wave cloud, the polarization pair (pol_u, pol_v) , and the interactions—are generated from this seed by the same local transition rule.

4.1 Internal addressing and W -islands

The internal non-spatial dimension has size

$$W = 3L^2, \quad (2)$$

where L is the lattice side (Sect. 3). During the Platonic Initialization, W is partitioned axiomatically into N_I identical W -islands, each containing

$$\ell = \frac{W}{N_I} \quad (3)$$

positions. Both integers are declared inputs of the seed—evaluated once at initialization and reported with the run—and no rule derives either of them from the dynamics. A W -island is an internal charge family: its ℓ copies are ℓ distinct W -addresses that share the same six charge bits but have distinct pB/sB patterns. Each such address maps to a distinct spatial position in the 3-torus—i.e., each copy is one bubble at one lattice site—so that the W -island is an abstract grouping, not a spatial substructure. The distinct polarizations provide the internal angular diversity needed for the Hofer effect. The detailed dynamics of how these copies aggregate into $1K + nD$ islands, and how the island population stabilizes, is treated in Section 8.

4.2 The Platonic seed

Among the many possible choices, we select the following deterministic Platonic seed:

- The spatial address $\mathbf{x}$ is set to the appropriate Cartesian positive coordinates $(0 \dots L - 1)$. Every source center is placed at the single lattice center $\mathbf{c} = ((L - 1)/2, (L - 1)/2, (L - 1)/2)$, so that all bubbles are born superposed at one point with zero radius—deterministically and without random numbers.
- The internal address w is immutable; charge bits are derived from its charge-family index $\nu = \lfloor w/\ell \rfloor$, so copies of a family have equal charges. In the seed used for the numerical experiments, $w_0 = \nu \bmod 2$, $w_1 = \lfloor \nu/2 \rfloor \bmod 2$, $q = w_1 \oplus w_0$ and color $= \nu \bmod 8$.
- The affinity a and the auxiliary leader identity $leader_w$ are initialized to the island index of the W -island to which w belongs (Sect. 4.1). A true chief (K) emerges later through encounter, as described in Section 8.
- The housekeeping counter k and the wavefront counter t are set to zero; the contraction/phase vectors $\mathbf{c}$ are zero and $s2B$ is reset.
- The momentum vector $\mathbf{m}$ is initialized to $\mathbf{0}$. The dynamic election path may install a direction at an expansion limit, once on the frame edge. Once a pair is an affinity-bound drive pair with nonzero $\mathbf{m}$, its transport direction is preserved rather than replaced at every era. Prepared-island probes explicitly seed that direction to isolate the inertial mechanism.
- The consumable translation impulse $reloc$ is zero.
- The wavefront data are seeded only at the source centers: $u = v = 0$ and $r^2 = 0$, with a fixed amplitude U_0 ; all other cells have $r^2 = \text{INF}$ and $u = v = 0$.

Real numbers and high-level functions are used only as intermediaries to compute these integer parameters; they are not stored in the lattice. The dynamics is therefore strictly integer-based.

4.3 What the seed does not contain

The seed contains no precomputed radius, no wave cloud, no $(\text{pol}_u, \text{pol}_v)$ pair, no $pB/sB/s2B$ pattern, no non-zero momentum $\mathbf{m}$, no interactions, and no randomized data. Those are the first emergent consequences of the seed and are described in Sect. 5.

4.4 The initial thermal bath

Dynamic initialization and evolution proceed concurrently under the same transition rules, with no global barrier between them. Their dependencies are local: one region may already sustain organized aggregates while another remains in a transient regime, and subsequent interactions may disturb previously stabilized variables.

Before the distance, polarization and island patterns settle into their operating regime, the system undergoes a preparation transient. Its proposed mixing role is a deterministic “thermal bath”; homogeneity and loss of dependence on the seed must be measured, not inferred from irregular motion alone. Statistics intended to characterize inertia must exclude an explicitly reported preparation window and compare successive measurement windows. A discarded prefix alone is not evidence of equilibration.

5 Phase step and emergent wave fields

Once the Platonic seed is fixed, the first CA ticks generate the wavefront geometry that feeds the frame FSM of Sect. 6. The subsections below describe those emergent fields.

5.1 Dynamic radius and the Euclidean metric

The wave field is seeded at the source centers, all of which are born superposed at the lattice center. From its source, each wavefront expands as a sphere in the cubic lattice, so we need a local, integer-valued approximation of the Euclidean distance from every cell to its source.

Squared-distance update. Let the source have integer coordinates (c_x, c_y, c_z) and let (x, y, z) be the cell coordinates, all taken modulo L along each axis of the 3-torus. The exact squared Euclidean distance is

$$r^2 = (x - c_x)^2 + (y - c_y)^2 + (z - c_z)^2.$$

In the 6-neighborhood (axis-aligned moves only), an outward step along the axis with absolute offset a from the source changes r^2 by

$$\Delta r^2 = (a + 1)^2 - a^2 = 2a + 1.$$

Therefore the local update is purely additive:

$$r_{\text{new}}^2 = r_{\text{old}}^2 + 2a + 1,$$

where a is the cell's own absolute coordinate along the moved axis. Each cell needs only its stored r^2 and its absolute offset from the source, both integers. The update is performed synchronously over the whole lattice at every tick: each cell writes its new value into a second, identical lattice (`grid` and `grid_next`), so the rule is a local double-buffer CA transition. In one tick a cell can only learn from its six immediate neighbors, so the reachable region grows by one lattice layer per tick, exactly as the information cone should—the cone of the maximum speed of information transmission v_{max} ; before the wavefront reaches a cell, $r^2 = \text{INF}$. Because the edge weights are non-negative, repeated synchronous sweeps converge to the same minima a Dijkstra/BFS relaxation would produce, but without any global queue or frontier list.

Radius correction method Several physical formulas require the ordinary radius r rather than its square: the wave envelope $\sin(r)/r$ and the radial shell count used for the $r \cdot \sin(r)$ density use r directly, while the wave phase $f = \text{effective_t}(t)$ is the triangular breathing phase of the local tick t (Sect. 6)—it rises one cell per frame from 0 to $L/2$ and falls back to 0, so over one era it selects the cloud point density over exactly the first lobe, from 0 to π and back. The topological constants $2R$ and `phase_full` remain in use for the polarization reconstruction of Sect. 5.3. The lattice is integer-valued, so r is the largest integer not exceeding the true Euclidean distance, $r = \lfloor \sqrt{r^2} \rfloor$. We propagate r alongside r^2 . When a cell with radius r passes the wavefront to a 6-neighbor, the neighbor's squared distance increases by $2a + 1$, which changes the true radius by 0 or 1. Hence the neighbor's integer radius is either r or $r + 1$, and the correct choice is the one that satisfies

$$(r')^2 \leq r_{\text{new}}^2 < ((r') + 1)^2.$$

This test uses only integer additions and comparisons. A one-step source-center motion changes any cell's radius by at most one, so a tiny correction loop around the previous r is enough to keep the value exact after the source moves. The discretization error is at most one lattice unit and does not accumulate, because r^2 is re-anchored by the source center each sweep and the radius is re-corrected from the exact r^2 .

Efficiency of the method Extracting r once per cell per tick would require a bounded loop of bit operations. By propagating r together with r^2 we replace that generic root extraction by a single local square-boundary test: the parent cell already knows its own radius, and the child radius is either the same or one larger. The source center re-anchors $r = 0$ each sweep, so errors cannot drift. This keeps the FSM transition small and avoids carrying a full integer square-root routine in the inner loop.

Why the 6-neighborhood? The 6-neighborhood corresponds to the three Cartesian axes and is the minimal set of edges compatible with the discrete rotation/reflection symmetries of the cubic lattice. Edge weights $(2a + 1)$ then give the correct squared Euclidean increments for those edges. Moves along face diagonals or body diagonals would require larger neighborhoods and non-integer corrections; the 6-neighborhood together with the squared-distance update already captures the Euclidean metric without adding extra edges.

Boundary behavior. Because the lattice is a 3-torus, the distance field wraps through the periodic boundaries. The update remains local: a step that crosses a face of the cube uses the periodic coordinate and the same $2a + 1$ increment. As long as the source center is taken modulo L in each axis, the resulting r^2 is the squared distance on the torus. Multiple source centers are handled independently, one per W -layer, and the local relaxation within each layer does not mix their fields.

5.2 The wave cloud

The wave cloud is not a precomputed mask. It emerges from a discrete wave field that propagates outward from each bubble. Each cell carries a wave amplitude that evolves according to an integer-only wave equation, while a pulsating spherical wavefront marks the cells whose propagated integer radius r equals the local frame radius $\text{effective_t}(t)$ (which advances one cell per frame at the maximum information speed $v_{\max}$) as *active*. The sieve bit $s2B$ is computed directly from the local displacement u : for $u > 0$,

$$((u \cdot t) \bmod \text{SHELL_TARGET}) < u, \quad \text{SHELL_TARGET} = 16384,$$

sets a trigger, and the result stored in the cell is $s2B = \text{active} \wedge \text{trigger}$. The bit is not carried forward between frames: each phase step recomputes it from the local u and t , so what the cell stores is always the test of the current step. The coincidence of *active* cells whose $s2B$ bit is true forms the sieve. Because the number of cells at radius r on a cubic lattice scales approximately as r^2 , the radial density of active cells with $s2B$ true follows $r \cdot \sin(r)$, which is the electromagnetic coupling density (see Figure 2). The underlying wave amplitude per cell is $\sin(r)/r$, so the $r \cdot \sin(r)$ density is a geometric consequence of binning active $s2B$ cells over spherical shells. The $s2B$ bit propagates this test between bubbles. The cloud is generated on the fly inside each light frame. At $t = 0$ the wave field is zero except for the excitations of fixed amplitude U_0 with $r^2 = 0$ at the source centers; all other cells have $r^2 = \text{INF}$ and $u = 0$. The constant U_0 is chosen independently of L (for example, $U_0 = 2048$ in the reference implementation), so the same seed remains valid as the lattice grows; in the $L \rightarrow \infty$ limit it becomes a point excitation on an unbounded lattice. As the wavefront radius expands and contracts over an era, one cell per frame (at the maximum information speed $v_{\max}$), the normalized wave amplitude yields the $\sin(r)/r$ envelope per cell, and the active-triggered cells form the $r \cdot \sin(r)$ density. The wave phase f , the Bresenham accumulator, and the active flag are therefore dynamical, not imprinted by fiat. The constants $U_0 = 2048$ and the sieve modulus $S = 16384$ are *free* parameters: nothing in the model fixes them; they are scale choices of the reference implementation. Their role is now quantified: U_0 sets the amplitude scale u on the active shell, and the electroweak channel opens or closes with the ratio u/S . The reference values were chosen so that the sieve is nearly dormant ($u/S \lesssim 1\%$), i.e. so that the *weak* channel is genuinely rare at the reference configuration; the choice $S = 2^{14}$, $U_0 = 2^{11}$ also keeps the modulo arithmetic in the cheap integer range. A principled derivation of both constants from the model is an open problem (Sect. 9).

5.3 Emergent polarization pair ($\text{pol}_u, \text{pol}_v$)

Each active cell carries an integer pair (u, v) , stored as pol_u and pol_v , that represents the transverse polarization in the plane tangent to the spherical wavefront. The pair is not a geometric function of the local radius; it is derived from the *broadcasted arrival stamp* $b(x)$ of an *elected momentum vector*. At every turn-around of the breathing wavefront the automaton elects one direction out of its own state, a helical walker broadcasts the news of that election across the layer, and every cell converts its private arrival stamp into a polarization phase. The mechanism has three stages—election, broadcast, and reconstruction—and uses only addition, subtraction, shift, and comparison.

Overview. The goal is to assign to each cell a direction in the plane perpendicular to an elected axis $\mathbf{m}$, using only integer arithmetic. The direction is encoded as a pair (u, v) lying on a circle of radius R^2 , where R is the emergent bubble radius. The angle $\theta = \text{atan2}(v, u)$ advances from 0 to 2π as the broadcasted arrival time grows, winding a polarization helix around $\mathbf{m}$. The automaton never computes θ explicitly; it computes (u, v) directly from the arrival stamp using integer square roots.

Stage 1: Election. Every cell owns a predefined value $\nu \in [0, N_I - 1]$ —its W -island index, fixed once by the internal address. At the expansion limit of the pulse (the exact tick where the wave turns from ascending to descending), every active shell cell is seeded with a *payload* that combines a score (derived from ν and a global seed s) with the cell’s linear index. Because the index occupies the low bits, the payload order is a strict total order: ties are impossible, so the winner is unique and independent of any scan order.

Each pass lets every cell adopt the greatest payload among its six face neighbors:

$$\text{payload}(x) \leftarrow \max(\text{payload}(x), \text{payload}(x \pm e_i)),$$

with alternating sweep directions; $2R + 8$ passes guarantee full coverage of the lattice. The true winner—the *seeded* cell holding the greatest payload—is identified once, at sowing time, while the payloads are still

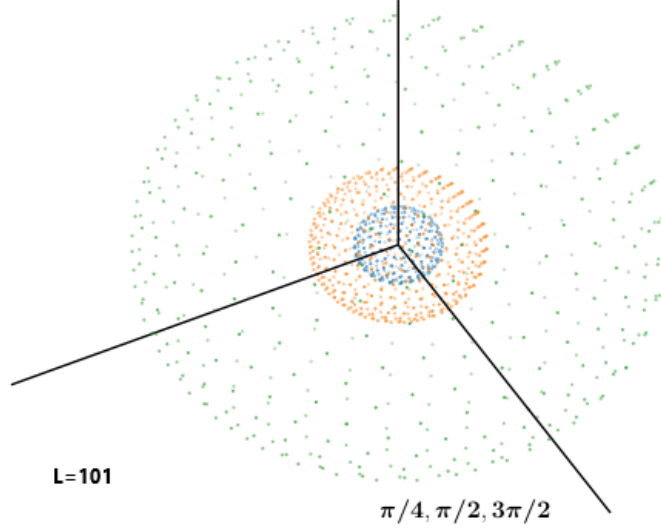


Figure 2: The emergent $r \cdot \sin(r)$ wave cloud. The radial interaction density of the active $s2B$ cells plotted against the radial distance r from the source center (horizontal axis: integer shell radius $r = 1, 2, \dots$ in cells; vertical axis: normalized density of active cells on the shell). The cell-by-cell wave amplitude follows $\sin(r)/r$, while the number of cells per spherical shell scales as r^2 ; their product gives the radial interaction density $r \cdot \sin(r)$ used for electromagnetic coupling. The curve is the analytical envelope $r \sin(r)$ (arbitrary vertical scale) overlaid on the simulated shell counts; the measured match is reported in the full manuscript.

unique; the diffusion flood merely transports the news and never redefines the winner. The displacement from the lattice center to the winning cell becomes the momentum vector $\mathbf{m}$: it is rescaled to $|\mathbf{m}| = L/2$ and written onto the source-center cell of the layer. That write is the sole dynamic initialization of $\mathbf{m}$; thereafter the inertia path (Sect. 6.1.2) treats $\mathbf{m}$ as immutable and only reads it to accumulate *reloc* impulses. The same elected axis drives the helical walker of Stage 2. The next expansion limit may elect a fresh direction; nothing is prescribed at the Platonic seed, and the orientation is read off the automaton’s own ledger at the instant the wave reaches its limit.

Stage 2: Broadcast. As soon as $\mathbf{m}$ exists, a helical walker traces the spiral arm on the pulsating bubble. It advances one cell per tick around the cylinder whose axis is the elected direction, keeping the radial error within about one cell by purely local comparisons among its six neighbors. The walker is an arbitrary-axis integer helix (implementation details in the source code: `spiral/spiral.c`). During the walk, each newly computed spiral point writes one plus the current tick into its own cell of a second value lattice, reserving $b(x) = 0$ for “never reached”.

Every other tick, with alternating sweep directions, every cell inspects only its six face neighbors and copies the greatest neighbor value whenever it exceeds its own:

$$b(x) \leftarrow \max(b(x), b(x \pm e_i)).$$

No cell ever senses beyond its neighborhood, yet these monotonic values diffuse as waves that in finite time reach every cell of the lattice—global coverage achieved purely by local moves, in lockstep with the automaton tick. Each cell therefore ends up holding the tick $b(x)$ at which the news of the elected momentum arrived.

Stage 3: Reconstruction. The arrival stamp $b(x)$ is the phase source. A full 2π turn is encoded by $\text{phase_full} = 2R^2$ units, where $R = L/2 - 2$ is the emergent bubble radius (R and phase_full are topological constants, Sect. 3.3). The cell phase is read directly from the broadcasted value:

$$\text{cell_phase} = (b(x) - 1) \bmod \text{phase_full}, \quad j = \left\lfloor \frac{\text{cell_phase}}{R} \right\rfloor.$$

The integer j ranges from 0 to $2R - 1$ and represents the position along the polarization circle. Writing $j_2 = j - R$ for $j \geq R$, the integer pair is reconstructed by:

$$\text{if } j < R: \quad u = R(R - 2j), \quad v = 2R \text{isqrt}(j(R - j)), \quad (4)$$

$$\text{if } j \geq R: \quad u = R(2j - 3R), \quad v = -2R \text{isqrt}(j_2(R - j_2)). \quad (5)$$

These formulas generate an integer pair (u, v) that lies on the integer lattice closest to the circle $u^2 + v^2 = R^4$ of radius R^2 (the equality is exact when the integer square root is exact, and is otherwise the closest integer approximation). The polarization angle, defined geometrically as $\theta = \text{atan2}(v, u)$, advances from 0 to 2π as the broadcasted value grows, winding the polarization helix around the elected momentum axis. This angle is never numerically evaluated by the automaton: it is the continuous geometric interpretation of the integer pair (u, v) , which encodes the direction implicitly. No trigonometric tables, no precomputed spiral, and no per-cell fixed constants are used: the direction emerges from the tournament over the W -ledger, and the phase from the integer square-root relation applied to the arrival time. Per the purity constraints (Sect. 3.3), the runtime performs no trigonometric computation whatsoever; atan2 is introduced here only as a descriptor linking the integer pair to its emergent angular variable.

Fidelity of the reconstruction. Two quantitative statements characterize how Eqs. (4)–(5) track the circle $u^2 + v^2 = R^4$ of radius R^2 . *Radial deviation.* The reconstructed pair never leaves the circle interior, and its deficit is purely the integer-square truncation: $u^2 + v^2 = R^4 - 4R^2\varepsilon$ with $\varepsilon = j(R - j) - \text{isqrt}(j(R - j))^2 \leq R$, so the radius deviates from R^2 by at most $2R$ integer units (relative deviation $\leq 2/R$, vanishing wherever the integer square root is exact). Measured maxima of the relative radial deviation decrease as $1/R$: 25.5% at $R = 3$, 11.6% at $R = 16$, 3.1% at $R = 64$, 0.39% at $R = 512$. *Angular winding.* The angle $\theta = \text{atan2}(v, u)$ winds monotonically from 0 to 2π as the broadcasted value grows over the $2R$ steps, but the per-step spacing is not uniform: because the formula parametrizes the circle linearly in j rather than in angle, the deviation from the uniform grid $\pi j/R$ saturates at about 19° (maximum; mean about 12°) for $R \gtrsim 16$, with special radii (e.g. $R = 4$) where the sampled angles coincide exactly with the uniform grid. Both statements are exact properties of the integer formula — the automaton itself evaluates no trigonometric function (Sect. 3.3) — and Table 2 tabulates them (program `polar_fidelity.cpp`, archived with the source).

Table 2: Angular and radial fidelity of the integer polarization reconstruction (Eqs. (4)–(5)) as a function of the emergent radius $R = R_{\text{MAX}} - 2$ (the tested lattice sequence $L = 7, 9, 11, 15, 19, 21$ corresponds to $R = 1, 2, 3, 5, 7, 8$). “Max $|\Delta\theta|$ ” is the maximum over the $2R$ steps j of $|\text{wrap}(\text{atan2}(v, u) - \pi j/R)|$; “mean $|\Delta\theta|$ ” its average. “Max rel. radial dev.”, is $\max_j (R^2 - \sqrt{u^2 + v^2})/R^2$, the largest deviation of the reconstructed radius from the true circle of radius R^2 ; it is bounded by $2/R$ and vanishes when the integer square root is exact. Angles are post-hoc diagnostics only: the automaton performs no trigonometric computation (Sect. 3.3). Program `polar_fidelity.cpp`.

emergent radius R	steps $2R$	max $ \Delta\theta $ (deg)	mean $ \Delta\theta $ (deg)	max rel. radial dev.
1	2	0.000	0.000	0.000
2	4	0.000	0.000	0.000
3	6	3.435	2.290	0.255
5	10	17.130	8.438	0.175
7	14	12.946	8.099	0.131
8	16	11.310	6.641	0.209
16	32	17.306	9.654	0.116
32	64	17.820	11.096	0.060
64	128	18.686	11.778	0.031

Once these fields are present, the cyclic evolution described in Sect. 6 begins. Within that structured loop particles move, interact, and display emergent physical behavior.

6 The light frame

Era and interaction timescales. An *era* is one complete expansion of a wavefront from zero to its maximum radius followed by contraction back to zero. A light frame is an internal update sequence, not an

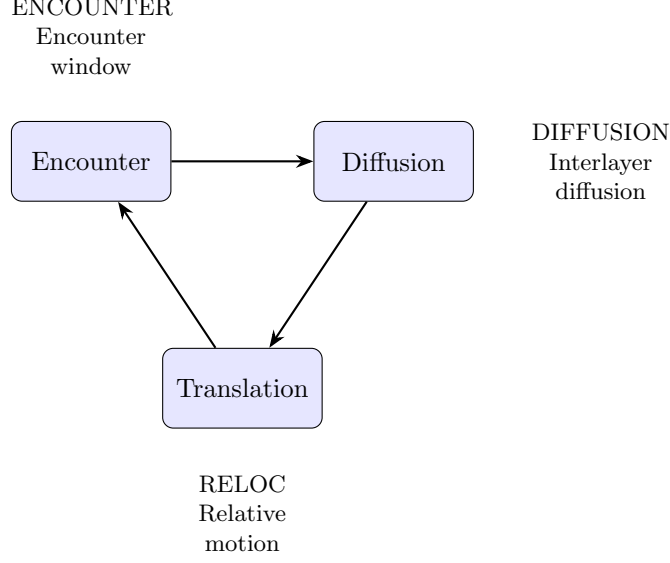


Figure 3: The light frame—the period of the information clock, within which a wavefront advances exactly one lattice cell at the maximum information speed $v_{\max}$ —has three stages: Encounter, Diffusion and Translation, delimited by the constants ENCOUNTER, DIFFUSION and RELOC.

era. Inertial interactions and the associated re-emissions are intended to occur within a few housekeeping ticks k , occupying only a small fraction of an era; they do not require completion of the expansion-and-contraction cycle. Dynamic initialization may require many eras before its variables stabilize. These timescales must be distinguished when interpreting numerical measurements. The light frame (Figure 3) is one complete light step of the cellular automaton. It is divided into five stages: **ENCOUNTER** (interactions and pair formation), **DIFFUSION** (propagation of contact, collapse, and translation information), **RELOC** (physical displacement of source centers), **REISSUE** (re-emission of the wavefront from the new source center), and **FLOOD** (final synchronization of the three lattices). The global housekeeping counter k advances synchronously in every cell, marking the current stage; the wavefront counter t is local and resets to zero whenever a bubble is reissued. The separation between the common k and the local t is what lets the finite-state machine coordinate essentially non-local operations while keeping them intended to be strictly non-signaling: no information and no influence ever travels faster than the maximum information speed $v_{\max}$ in the idealised local rule, and causality is protected. For the host-side scheduler of the present implementation this remains a design goal, not an established property. Despite its name, the light frame is not a unit of light propagation: it is the period of the information clock, within which a wavefront advances exactly one lattice cell, i.e. at the maximum speed of information transmission $v_{\max}$; this rate defines the information clock rather than being an emergent prediction (the corresponding fidelity checks are reported in the full manuscript). The speed of light does not appear anywhere in the transition rule; it is emergent. An observer that measures distances and times by a round-trip convention reads the photon of Sect. 6.1—a pair of bubbles riding the wavefront—as propagating at this speed and names it c . In particular, the **ENCOUNTER** stage is superluminal in coordinate terms (it compares the whole lattice along W in a single sweep), but it is intended to be strictly non-signaling and cannot be manipulated by any observer; causality is thereby protected in the idealised rule, with the same host-side caveat as above. **Definitions.** The frame constants of the light frame are the following:

- **DIAG**
 $\lfloor \sqrt{3} L \rfloor$: diagonal length of the lattice; upper bound for interaction distances.
- **RMAX**
 $L/2$: maximum bubble radius before wrapping (the breathing-phase amplitude).
- **ENCOUNTER**
 $2W$: encounter window.
- **DIFFUSION**

$ENCOUNTER + 2W + \lfloor \sqrt{3}(L-1)/2 \rfloor + 9(L-1) + 2W$: diffusion range along W ; also alerts all layers of an imminent collapse.

- **RELOC**

$DIFFUSION + 3(L-1)$: translation phase.

- **REISSUE**

$RELOC + 1$: re-emission of the wavefront from the new source center.

- **FLOOD**

$REISSUE + 3(L-1)$: final synchronization of the three lattices.

- **FRAME**

Same value as $FLOOD$; total duration of the light step.

- **effective.t**

Triangular breathing phase (the active-shell radius): $\text{effective.t}(t) = t \bmod L$ for $t \bmod L \leq L/2$, and $L - (t \bmod L)$ otherwise. The active-shell radius therefore advances one cell per step of the local tick t —i.e., per light frame—from 0 up to the amplitude $L/2 = RMAX$ on the expansion branch and back to 0 on the contraction branch; each branch lasts $L/2$ frames, so one era (one full expansion-and-contraction cycle) has period $L = 2RMAX$ in t and spans L light frames. An era is distinct from a light frame and from the housekeeping tick k .

6.1 Bubble, pair, singleton, and cluster

A *bubble* is an expanding spherical wavefront of information organized across cells within a common layer. Spherical propagation is determined by comparing the dynamic radius r to the wavefront tick t .

6.1.1 Pair

Two bubbles are overlapping if they share the same spatial coordinates and $t_1 = t_2$. Overlapping bubbles can form a *pair* if they have the same affinity, that is, $a_1 = a_2$; the two partners share a center and phase, and their charges are complementary or matching depending on the pair's role. Pairs may remain bound to a particle as *dressings* pairs or propagate freely as photons; a free pair expands outward and is gradually consumed, releasing two singletons at distinct locations.

6.1.2 drive pair

A pair can act as a *drive pair*, repositioning other bubbles and providing inertia. A drive pair is a reciprocal photon-type pair that has adopted the island's affinity a and carries a non-zero transport vector $\mathbf{m}$, typically on the scale $L/2$. Neither a nonzero vector alone nor receiving a translation impulse makes a source a drive pair. A free pair must first acquire the particle's affinity. Consider an island of N equal-charge elements, one chief K and $N-1$ delegates D , all sharing a . Its $K \times D$ and $D \times D$ contacts maintain dynamic cohesion. Without drive pairs its center of mass remains at rest in the lattice frame, although its constituents may rearrange. Bound pairs supply persistent internal propulsion through $P \times K$ and $P \times D$ contacts. A rightward resultant of their transport vectors drives a rightward mean displacement of the cohesive island. Few aligned pairs give sparse kicks and low velocity; many nearly collinear pairs give more frequent coherent kicks and higher velocity, up to the lattice transport bound. This is the proposed mechanism of inertia, not an externally imposed velocity of the aggregate. The reference simulation algorithm resolves each unordered source contact once per light frame, counting a reciprocal pair once rather than once per half or overlapping voxel. A pair can deliver at most one unit face-step to one contacted body element, in a direction selected from $\mathbf{m}$; receiving that step preserves the target's K/D identity. $\mathbf{m}$ is not a displacement of $L/2$ cells and is retained while the pair remains a drive pair. These mechanical, same-affinity contacts are not gated by the electroweak sieve.

6.1.3 Singleton

An unpaired bubble is called a *singleton*; it is not a generic synonym for bubble.

6.1.4 cluster

The *cluster* is a group of superposed pairs having the same affinity, same true bB bits and the same (active) pBits.

6.1.5 Specialized pairs

Certain pair combinations are not allowed to form clusters:

- Neutrino and antineutrino pairs possess color values N, N and $\bar{N}, \bar{N}$, respectively.
- The up quark fragments u exhibit equal, nontrivial colors and a positive charge in the case of Orbis.
- The drive pair adds to the relativistic mass of the propelled fermion.¹
- Pairs with fully complementary charges are *gravitons*. They operate in both Orbis and Umbra,² have fixed oscillation frequency 2, and mediate gravity as a cross-sector, charge-sign-independent attraction; without them, Umbra matter would appear inert in Orbis. Electromagnetic static forces are instead mediated by sectorial photonic pairs (Sect. 7.1.2). Graviton-induced gravity is detailed in the full manuscript.

6.2 Multi frequency mechanics

Dynamic initialization (the static sine/momentum masks having been removed) generates the half-wave pattern for the emergent density $r \cdot \sin(r)$, the polarization pair ($\text{pol}_u, \text{pol}_v$), and the active wavefront from the integer wave equation and the elected-axis broadcast alone (Sect. 5.2 and Sect. 5.3). The radial density of active cells that pass the sieve therefore follows $r \cdot \sin(r)$ as a geometric consequence of the $\sin(r)/r$ amplitude per cell times the r^2 shell volume; no precomputed mask is required.

A photon is a single cluster with all pairs superposed. Frequency is not a continuous parameter but an integer count: a single overlapping pair at the same lattice site and wavefront radius has the fundamental even frequency 2; a stack of n such pairs has frequency $2n$. A multifrequency photon is a superposition of several such stacks (or a cluster containing populations at different even frequencies 2, 4, 6, ...). There are no continuous harmonics; the spectrum is discrete and composed only of these even-integer modes.³

Clusters are therefore intrinsically multifrequency. Each constituent pair (or stack) carries its own integer population (`pair_count`). The automaton reads frequency as a population, not as a waveform. In the reference dynamics the count is stamped once at formation and spent by the turnaround consumption; stacks grow only where a harness seeds them, because pair formation requires two sources in the single state S and cannot re-reach a formed pair. The candidate absorption rule `PAIR_STACK_ABSORB_FSM` (off in the reference build) makes each detected formation register the population of identical superposed stacks on the new pair, so a stack of n accumulates toward $n + 1$ per encounter; dynamic stacking therefore remains part of the required mechanics below.

Limitation of the present mechanism. The current implementation does *not* yet determine the interaction point inside the $r \cdot \sin(r)$ cloud from the frequency. The active shell is set solely by the local wavefront tick of the sources ($r = \text{effective.t}(t)$), and the sieve $s2B$ is the local test

$$((u \cdot t) \bmod S) < u, \quad S = \text{SHELL_TARGET},$$

gated by that same active flag. Consequently a multifrequency cluster interacts wherever its current active shell happens to lie; there is no selection of a radial phase that corresponds to a particular mode of the discrete spectrum. Electromagnetic (and magnetic) contacts therefore cannot be localized to the correct point of the cloud for a given frequency. Moreover, dynamic initialization elects the direction of the momentum vector $\mathbf{m}$ and installs it on the source center, but it does *not* yet materialize a straight-line sequence of cells along $\mathbf{m}$ on the lattice (the polarization walker produces a spiral, not a ray).

¹This mirrors a Kantian view: particles do not self-move but require an external cause—here, interaction with fragments—to change their state.

²Orbis and Umbra coexist spatially.

³Laboratory and environmental photons are groups of such clusters, combined in time and space.

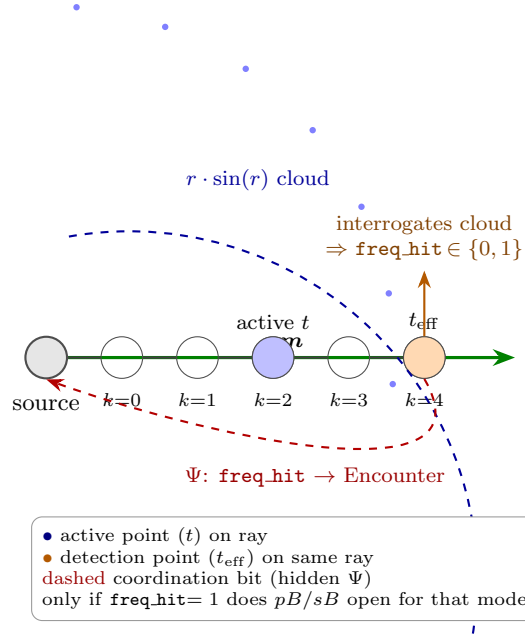


Figure 4: Multifrequency selection on the lattice. Dynamic initialization stamps a pure-climb ray along the elected momentum $\mathbf{m}$. The ordinary active shell marks one cell of the ray (tick t). Independently, t_{eff} derived from `pair_count` selects another cell of the same ray; that cell interrogates the already-propagated $r \cdot \sin(r)$ cloud and yields the binary coordination bit `freq_hit`. The bit is written into source-center Ψ and, by the end of Encounter, gates the electromagnetic channel for that discrete mode. Different modes therefore interact at different radial phases of the same cloud.

Required mechanics. Fixing multifrequency requires a linear sequence on the lattice together with a binary, frequency-conditioned trigger (Fig. 4). The construction must remain pure (only addition, subtraction, bit shifts and comparisons of dynamic operands).

1. **Linear sequence along $\mathbf{m}$.** At the moment the axis is elected and $\mathbf{m}$ is written on the source center, a pure-climb Bresenham walk (the climb component of the existing polarization DDA, with orbital budget set to zero) stamps successive cells along the ray with the step index $k = 0, 1, 2, \dots, R_{\text{MAX}}$. Only increments and the already-supported DDA steps are used; no multiplication or division of dynamic quantities occurs. The stamp may reuse `bstamp` or a short dedicated field. Cost is $O(R_{\text{MAX}})$ and is incurred only on axis election.
2. **Effective tick from pair count.** The first action of Encounter for any source that carries a non-trivial `pair_count` is to form

$$t_{\text{eff}} = t_{\text{eff}}(\text{pair_count})$$

by shifts and additions alone (for example $t_{\text{eff}} = \text{pair_count}$ or $t_{\text{eff}} = \text{pair_count} \ll 1$). The mapping remains a free scale choice, but it is deterministic and local to the source data.

3. **Two points on the same line.** On the stamped ray there are two distinguished cells:
 - the *active point*, corresponding to the ordinary local tick t (intersection of the current active shell with the ray, or the cell whose stamp equals t);
 - the *detection point*, the cell whose stamp equals t_{eff} .

The detection point performs a purely local interrogation of the wave cloud already present on that cell (for example $\text{active} \wedge s2B$, or a positive amplitude u). The outcome is a single binary result

$$\text{freq_hit} \in \{0, 1\}.$$

4. **Superluminal propagation of the bit.** The bit `freq_hit` must become available by the end of Encounter as the effective trigger of the electromagnetic (or magnetic) channel. It is written into the

draft of the source-center cell (or reduced by a short OR-sweep strictly along the already-marked ray). Because the write targets a coordination field and does not alter readable physical fields of distant cells, the propagation is of the same character as the other non-local updates already performed inside Encounter.

Only after `freq_hit` is true is the ordinary pB/sB channel allowed to fire for that frequency mode. Different modes of a multifrequency cluster therefore interact at different radial phases of the same cloud; the sieve $s2B$ continues to gate local amplitude, but only after frequency has first fixed *where* on the ray the gate is examined.

Causal coherence (superluminal non-signaling). The construction is examined against the formal no-signaling argument of the full manuscript. That argument partitions each cell into readable physical fields Φ (charge, affinity, radius, wave amplitudes, polarization bits, phase, active flag, local clock t) and hidden coordination fields Ψ (housekeeping tick, W -ledger including `pair_count`, source kind, relocation impulses, ...). A minimal observer reads only Φ and acts only by local rules; physical effects of any change in Ψ reach other cells solely through the local six-neighbour transition and therefore at speed at most $v_{\max}$.

The bit `freq_hit` remains coherent with that partition if and only if the following conditions hold:

- `freq_hit` is treated as a coordination quantity (Ψ), never as a readable physical field (Φ). A minimal observer cannot sense it directly.
- Its only non-local write is into the draft of the source-center cell (or another already-authorized Ψ location of Encounter). Intermediate cells of the ray are not given new readable state.
- The physical consequence of the bit—the eventual firing of the pB/sB channel—is realized exclusively by the ordinary local transition rules (collapse, reissue, one-step motion, ...). From that moment onward every influence propagates at or below $v_{\max}$.
- No admissible local intervention by an observer can modulate the bit in a controllable, spacelike fashion: t_{eff} is computed from data already resident at the source, and the cloud interrogation is a passive read of a field that has already been propagated by the local wave rule.

Under these restrictions the bit introduces no new species of superluminality; it re-uses the same non-signaling license already granted to Encounter. The host-side scheduler of the present implementation inherits the same design-goal caveat already stated for all coordinated updates (see the full manuscript): the property is required of the idealised transition function, not yet established for the concrete scheduler.

The dynamic half-wave generation, the `pair_count` field with its formation and turnaround-consumption paths, the election of $\mathbf{m}$ and the amplitude-dependent sieve are already present in the executable (`initSim.cpp`, `simulation.cpp` `phase_step`, `interaction.cpp`, `polarization.cpp`). Stack growth is not part of the reference dynamics: the formation rule requires both sources in the single state S , so the count is stamped once at formation and spent by consumption, and multifrequency populations exist only where a harness seeds them; the candidate absorption rule `PAIR_STACK_ABSORB_FSM` (compiled off in the reference build) grows identical superposed stacks by one unit per encounter. The missing pieces are the pure-climb linear stamp along $\mathbf{m}$, the derivation of t_{eff} , the binary detection at the stamped cell, and the coordination-only propagation of `freq_hit`; all four must be introduced before electromagnetic contacts can be frequency-correct.

6.3 Interaction principles

The guiding principles are: **Wavefronts clash.** The wavefronts of both bubbles must be active, i.e. $r_1 = \text{effective_t}(t_1)$ and $r_2 = \text{effective_t}(t_2)$, where r_i is the propagated integer radius of bubble i and t_i is its local frame counter. This condition is assumed in all rules below.

Aggregation and dissolution. Bubbles aggregate by sharing a common affinity; the smallest value prevails, $a_1 = a_2 = \min(a_1, a_2)$. On dissolution, they revert to their layer indices, $a_1 = w_1$ and $a_2 = w_2$. If a singleton meets a pair, the pair takes the singleton's affinity, $a_2^1 = a_2^2 = a_1$. The value $a = W$ marks an orphan. The idempotent min operator naturally supports an ontological model of entanglement.

Charge conjugation. Charge conjugation constrains interactions by requiring charge neutrality in the outcome. Electric neutrality requires a pair of opposite charges; a single bit cannot be neutral alone.

Weak neutrality follows from the combination of the two weak bits in both *Orbis* and *Umbra*. Strong neutrality is 000 (or 111 for anti-neutrality); a pair with complementary color bits (e.g. 101 and 010) is also neutral. In general, interactions drive the resulting charge combination toward neutrality.

Reciprocity. Because only one draft can be updated at a time, the rules must later update the counterpart. This reciprocity is essential for the next subsection.

6.4 Encounter

Encounter is the first step of the light frame (ending at *ENCOUNTER*) and contains the main interaction rules. Each site in the main lattice is compared with its partner along the W dimension. This partner comparison is a rule-evaluation device that pairs lattice sites by internal address; it does not move bubbles along W —physical motion and merging always occur in the 3-torus (Sect. 4.1). For any frequency-bearing source the first actions follow Sect. 6.2: derive t_{eff} from **pair_count**, interrogate the cloud at the corresponding stamped cell of the linear sequence along $\mathbf{m}$, and obtain the binary coordination bit **freq_hit**. Only when that bit is set is the ordinary wave-phase f and the sieve $s2B$ allowed to open the pB/sB channel for that mode. Particles form or reform here. We postulate that the weak force only causes collapse, not limited reissues. Comparisons split into two cases: superposing or distinct bubbles.

6.4.1 Superposing bubbles

Two overlapping bubbles with the same wavefront tick $t_1 = t_2$ can form a pair if they satisfy the following conditions:

- $(q_a \oplus q_b) \wedge (w1_a \oplus w1_b) \wedge (w0_a \oplus w0_b) \wedge (c2_a \oplus c2_b) \wedge (c1_a \oplus c1_b) \wedge (c0_a \oplus c0_b)$
- $(q_a \oplus q_b) \wedge (w1_a = w1_b) \wedge (w0_a \oplus w0_b) \wedge (c2_a \oplus c2_b) \wedge (c1_a \oplus c1_b) \wedge (c0_a \oplus c0_b)$
- $ch_a = ch_b = 000000B$
- $ch_a = ch_b = 111111B$
- $(q_a = q_b = 0) \wedge (w1_a = w1_b = 0) \wedge (w0_a = w0_b = 1) \wedge (color_a = color_b) \wedge (color_a \neq 000B \wedge color_a \neq 111B)$
- $(q_a = q_b = 1) \wedge (w1_a = w1_b = 1) \wedge (w0_a = w0_b = 0) \wedge (color_a = color_b) \wedge (color_a \neq 000B \wedge color_a \neq 111B)$

The bit order $w_1 w_0 q c_2 c_1 c_0$ and the color code follow Table 1. These rules implement the formation of pairs, as defined in Sect. 6.1. If pairs are detected, one frequency unit is registered on the pair stack (Sect. 6.2); the wave phase f itself remains the triangular breathing phase of the local tick. $s2B$ is updated as $s2B' = s2B$ **and active**. Notice that the pairs are formed in two steps: first, cell a compares with cell b , and later, cell b compares with cell a (reciprocity). The formation of clusters is made during this step too, also following the reciprocity principle. Pairs are allowed to superpose if they are equal and were formed by the two first rules above.

6.4.2 Distinct bubbles

The w_1 bit separates distinct-bubble interactions into same-sector ($w_1^1 = w_1^2$) and inter-sector cases; we first treat same-sector interactions. If two same-sector singletons have opposite charges and both kB bits false, we have annihilation, with both bubbles being reissued from the contact point and their affinity receiving their default values. The draft kB bit turns true, triggering a collapse. Else if two singletons have the same charge, we have fermion cohesion. One bubble is reissued from the contact point, while the other, from its active sB site. Also, they become entangled ($a_1 = a_2$). On the other hand, if the two bubbles share the same affinity, but have different charges, the bubble whose wavefront at the contact point has $pB = 1$ is reissued from the contact point. For bound $P \times K/D$ transport, the pushing pair's $\mathbf{m}$ selects a unit step accumulated in **reloc**, as specified in Sect. 6.5; copying the full magnitude of $\mathbf{m}$ as a displacement is not the inertial rule. The following contact-site reissue description concerns the other interaction channels. That is, one bubble is reissued from $\mathbf{x}1$, while the other from $(\mathbf{x}_1 - \mathbf{x}_2) \bmod L$. For color-neutral combinations there are two cases: gluon $\times$ gluon and quark $\times$ gluon. In both cases, the two bubbles are reissued from the contact point and the partners colors are exchanged. In the electroweak

sector, both *active* bits must be true; together with *s2B* they determine the harmonic behavior. This sieve mechanism applies only to electroweak interactions and helps explain why the strong force is stronger. The weak interaction occurs when the weak charges are combined to neutrality and either ($pB_1 = 1$, $pB_2 = 0$) or ($sB_1 = sB_2 = 1$); it always triggers collapse ($kB = 1$). The electric case uses one *pB* bit, the magnetic case one *sB* bit. If both *pB* bits are true (or both *sB* bits in the magnetic case), a collapse is triggered ($kB = 1$). If only one *pB* bit meets the other's surface, the interaction is adiabatic: no collapse occurs, but the bubbles exchange affinity and time and relocate to each other's center. The *a* and *t* values then spread throughout the layer. At the initial singularity, all bubbles overlap. If two bubbles belong to different sectors and $t = \lfloor \text{RMAX}/2 \rfloor$, they are reissued from their active *pB* and *sB* sites, dispersing the singularity and separating Orbis from Umbra. We call this *dispersion*. This timed channel is a candidate mechanism: it is compiled off in the reference build (candidate macro `HOMB_PRODUCER.FSM`), so the reference run separates the sectors through its ordinary encounter channels at first contact, and the dispersion itself is not yet established. Where it is compiled in, the integer floor pins the event to a shell of radius $\lfloor \text{RMAX}/2 \rfloor$, which is at least one cell for every admissible *L* (one cell at $L = 5, 7$; two at $L = 9$) and never the center.

6.4.3 Inter-sector interactions

For inter-sector interactions, fully complementary charges lead to *singularization*; otherwise the electroweak rules above apply. Singularization is the counterpart to dispersion. The bubbles are reissued from the contact point ($pB_{curr} = \text{true}$).

6.5 Diffusion, translation and collapse in one reading

Three coupled tasks close the frame—diffusion, translation, and collapse handling—and are described together because they share the same coordination state.

Diffusion propagates the information needed for translation, ending at the `DIFFUSION` tick (Figure 5). If any collapse bit *kB* is set, it spreads to all bubbles sharing the same affinity, identifying every contributor to the collapsing particle. The wave phase *f* is not diffused: it is the local triangular breathing phase $\text{effective.t}(t)$ of each cell (Sect. 6). Phases incremented by multiples of the era period interact with the wave-cloud sieve to form concentric bubbles; the active wavefront point ($t = d$) is forwarded to the Euclidean bubble through *s2B*. At the same time, the translation vector *c* diffuses through the layer: each cell copies the first non-trivial *c* found among its neighbors. Finally, *sB* homing increments *c* along any axis where a neighbor has *homB* = 1, but only when the local cell has *sB* = 0. By the end of diffusion, *c* is ready for translation. **Translation.** This part of the light step addresses bubble

translation itself, building on the diffusion step. It ends at the `RELOC` tick. Bubbles shift in specific directions (north, west, down) until they have completed their translation process, including the inertial transport mechanism. The example rule is: if $\mathbf{c}_x^{\text{north}} > 0$, copy the content of the neighbor cell to the draft cell and decrement $\mathbf{c}_x$ at the draft cell. Since wavefront propagation is guided by the dynamic radius *r* propagated from the source cell, when a bubble is re-emitted due to interaction with other bubbles, all pertinent cell information must be shifted, or relocated, to the new position determined by the variable *c*. This ensures that the bubbles are relatively displaced from each other. The final operation sets $t = 0$ at the center cell of the relocated pattern. Translation is the main consequence of an interaction. Bubbles are re-emitted either immediately after an interaction or through wrapping. When the bubble is reissued, the previous wavefront continues to propagate until it fades through wrapping. In this state, the bubble is referred to as an *orphan*, meaning it has affinity disabled but is still capable of non-collapsing interactions. In this case, $a = W$ for the outer cells, set during the translation step. To complete, the diffusion of the time variable *t* occurs here at the last tick: When a cell has a neighboring cell with a lower *t* value, it updates its own *t* variable to match the lower value. With this rule, the relocated bubble is capable of expanding again. This timing is crucial to avoid the 'clean diamond' pattern with the new values overrunning the orphaned data. When the translation is complete, bubble variables are prepared for the next cycle. If the time frame completes, the *k* counter resets to zero, preparing the system for the next iteration. At the conclusion of the process, all *c* vectors will have a null value. For internal inertial transport, source contacts are gathered during the encounter window and resolved at the end of the light frame. The numerical experiments use the following operational choices to evaluate the proposed transport mechanism. Reciprocal $K \times D$ and $D \times D$ face-steps reduce separations greater than one cell without changing the sum of body positions. Each body element can move at most once per frame. Each reciprocal drive pair pair can push at most one contacted body element; among

Diffusion Phase

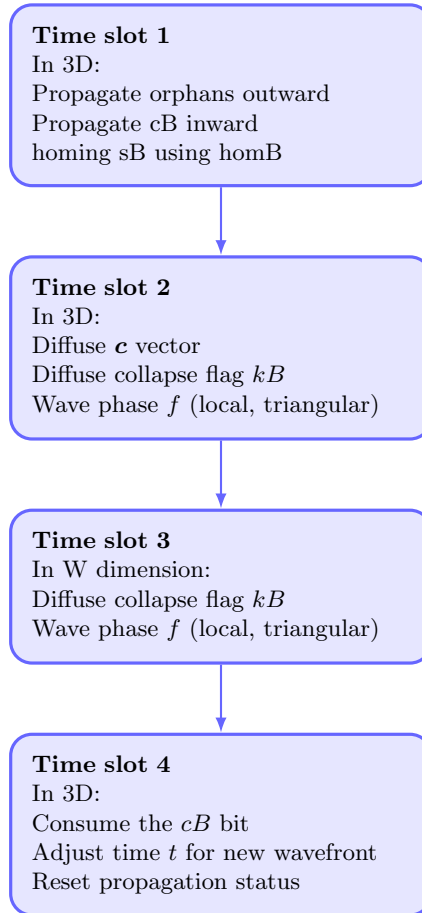


Figure 5: The Diffusion phase is composed of several time slots. It spreads all information necessary for translation.

eligible contacts it prefers the rear element along $\mathbf{m}$, with a rotating address tie-break. An integer DDA distributes successive face-steps in proportion to the absolute components of $\mathbf{m}$, retaining their signs. Zero-radius phases have no extended propelling surface. Both halves of the bound pair recycle by the same bounded face-step toward the contacted constituent. These are reference transport rules to test the proposed mechanism, not a derivation of Newtonian mechanics or rotational invariance. The resulting unit step, rather than the full vector $\mathbf{m}$, enters *reloc*. At the frame boundary, *applyMomentum* translates the bubble's layer state periodically, consumes the step and preserves its charge, affinity, source kind, pair identity, clock and transport vector. Translating the distance field together with its source avoids leaving a false source or stale distance minima. The source transaction commits after the voxel sweep so later cells cannot overwrite previously registered contacts or source changes. The partner rotation selects a W identity; active flags are read at the current physical instant rather than from the preceding frame's shell. The simulation algorithm uses globally coordinated source snapshots, contact arbitration, and layer translations. A strictly local cellular realization of that coordination remains to be established; the mechanical tests below validate the reference algorithm, not the purity constraints of Sect. 3.3. Dynamic initialization is never suspended, but election does not overwrite the direction of an already bound drive pair. Let N denote the number of equal K/D body elements, excluding the drive pair dressing. For n successful collinear unit kicks in a frame, with reciprocal cohesion contributing zero net displacement, the body's mean center of mass advances by

$$\Delta_{\text{CoM}} = \frac{n}{N} \quad (6)$$

cells per frame, with $n \leq N$. Oppositely directed steps enter with their signs; for general directions use $N^{-1} \sum_i \Delta \mathbf{x}_i$. This is a measured displacement identity, not a velocity assigned to the island or a mass law. drive pair-pair centers must be counted once each when separately measuring the dressed particle. Without propulsion, reciprocal cohesion preserves the body's center of mass. The displacements reported here are measured in cells per light frame. The housekeeping counter k indexes the internal stages of that frame and is not the physical time used in these measurements. Non-drive pair pairs behave differently: in $K \times K$ or $D \times D$ between different tribes the source centers move one light-step away from each other, exchanging momentum through *reloc*. An unbound pair may acquire an island's affinity through an admissible encounter. Subsequent same-affinity contacts provide propulsion; a pair of a different affinity cannot act as that island's internal drive pair. Receiving a displacement never promotes a K/D source to P . **Annihilation and collapse.** Annihilation and light-matter collapse both occur during the encounter stage. In annihilation, the affinity reverts to its default value $a = w$, and the resulting bubbles become raw material for new particles. This ontological collapse introduces an arrow of time at the fundamental level, reinforcing the second law of thermodynamics and pushing Poincaré recurrence far into the future (Sect. 9.1).

Reversibility. Because Fredkin and Margolus are cited as bases (Sect. 2), the reversibility status should be stated explicitly. The transition is *not* injective: (i) the *s2B* sieve discards information (the gate result is a single bit derived from a product modulo S); (ii) collapse sets *kB* and reverts affinity to its default, and reissue resets the local clock t ; (iii) relocation moves source centers and the flood stage synchronizes the three lattices by overwrite. Each of these steps is many-to-one, so the map from one light frame to the next is not reversible, and no inverse step is defined. This is a deliberate difference from Fredkin's reversible-computation program: the model is a dissipative deterministic automaton with an explicit arrow of time (see above), and it is not a candidate for Landauer-free computation. The claim is that the irreversible steps are confined to the hidden coordination fields and to the aggregate bookkeeping, so that the *physical* (readable) evolution keeps its local, causal character; reversibility of the whole map is neither assumed nor required.

6.6 Main lattice updating and partner refresh

At the end of each k tick, the draft lattice is transferred to main. If k lies inside the *ENCOUNTER* window, draft is first shifted along W to enable encounter. This shift is a cyclic permutation of internal addresses for evaluation, not propagation of bubbles along an extra spatial axis—all physical motion remains confined to the 3-torus (Sect. 4.1). The partner lattice is then refreshed from main so the next encounter can compare main and partner along W ; it copies the wave phase and active flag, $f_i^{\text{partner}} = f_i^{\text{main}}$ and $\text{active}_i^{\text{partner}} = \text{active}_i^{\text{main}}$.

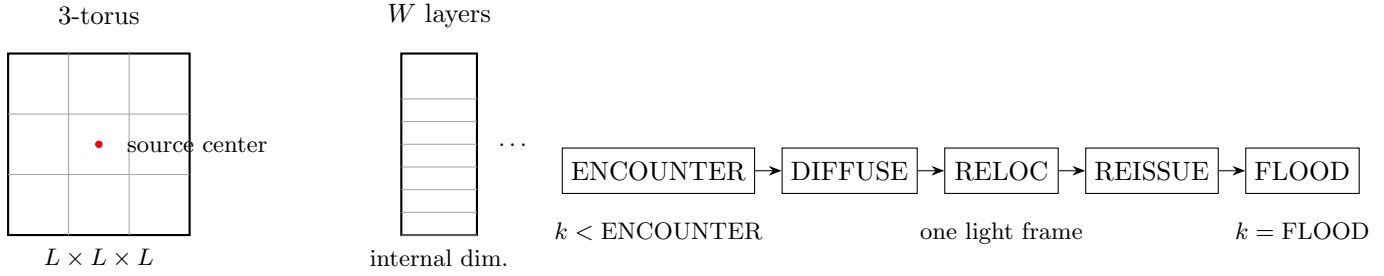


Figure 6: Model architecture. Left: the $L \times L \times L$ 3-torus (spatial fabric), with one source center per layer emitting a spherical wavefront. Middle: the internal dimension of W layers; each layer carries one bubble and the layers are compared by the encounter stage. Right: the light-frame pipeline of the finite-state machine—ENCOUNTER (interaction rules), diffusion of affinity and flags, relocation of sources, reissue, and flood—executed in FRAME housekeeping ticks k ; the wavefront advances exactly one cell per tick, at the maximum information speed $v_{\max}$.

6.7 Algorithmic summary of one light frame

For reproducibility, Algorithms 1–3 give a self-contained account of one complete light frame: Algorithm 1 is the frame driver (incremental distance field, integer wave update, the six interaction stages, the charge-conjugation hook, momentum and lattice swap), and Algorithms 2–3 give the interaction rules invoked by the ENCOUNTER stage. Together they define the full transition function; the repository (Ref. [10], `simulation.cpp`, `interaction.cpp`) is not needed to reproduce the dynamics. The notation follows Table 1: a charge word is written $w_1 w_0 q c_2 c_1 c_0$ (most significant first), $c_2 c_1 c_0$ is the color, 000B = N and 111B = $\bar{N}$, and $\bar{ca}$ denotes the six-bit complement of the charge word ca . Subscripts a and b label the two interacting bubbles, so q_a is bit q of bubble a , $w_{1,a}$ bit w_1 of bubble a , and so on. As given in Sect. 6.6, SWAPLATTICES transfers the draft to the main lattice at the end of every k tick; at $k = 0$ the partner is refreshed from main (so the next encounter compares main and partner along W), and while $k < \text{ENCOUNTER}$ the partner W -slices are rotated by ROTATEPARTNERS to schedule the cross-layer adjacency.

7 Interactions

Some interaction details are now clarified.

7.1 Charge driven interactions

7.1.1 Static interactions

Static long-range interactions are mediated by free pairs that ride the flux of orphans emitted by charged sources. The *species* of the mediating pair distinguishes the force: sectorial photonic pairs (sensitive to the electric bit q and to the polarization channel pB or sB) mediate electromagnetism; fully complementary pairs—gravitons—mediate gravity. In both cases the kinematic skeleton is the same: an orphan from source q_1 meets a free pair; both are reissued at the contact; if the pair later meets q_2 it is reissued there, acquires q_2 's affinity, and may become a drive pair of q_2 . The same process runs in reverse. What differs is which pair species is admitted and how the resulting drive-pair contacts bias the motion of the two sources.

7.1.2 Coulomb interaction

The proposed Coulomb mechanism uses *photonic* pairs, not gravitons. Orphan wavefronts from a charged source guide free photonic pairs toward another charged particle. A photonic pair becomes that particle's drive pair only after acquiring its affinity; subsequent internal contacts can then transport its K/D constituents in the pair's $\mathbf{m}$ direction. The receiving constituent does not itself become a drive pair. Same-sign electric charges are conjectured to bias those contacts so that the two sources are driven apart; opposite signs, so that they are driven together. The polarization bit pB opens the electric channel; the magnetic analogue uses sB (next subsubsection). The current internal-transport tests do not establish the

Algorithm 1 One complete light frame (the full per-cell transition function, one synchronous k -tick).

Require: $L, W = 3L^2$, $RMAX = L/2$, and the stage bounds $ENCOUNTER < GSLOT_Z < DIFFUSION < RELOC < REISSUE < FLOOD = FRAME$ (Sect. 6); the housekeeping tick k , the wavefront tick t , and the wave phase $f = \text{effective_t}(t)$ are the cell fields of Table 1

Ensure: the three lattices advanced by one light step

```

1: procedure LIGHTFRAME
2:   UPDATEPULSATINGWAVEFRONT  $\triangleright$  incremental distance field: radii  $r^2, r$  from the moving source
   center by  $\Delta=2a+1$  increments, no square roots
3:    $pulse\_tick \leftarrow pulse\_tick + 1$ 
4:   PHASESTEP  $\triangleright$  integer wave update  $v \leftarrow v + (\nabla^2 u \gg s)$ ,  $u \leftarrow u + v$ , damping  $v \leftarrow v - (v \gg s_d)$ ,
   shifts  $s, s_d$  radius-dependent; hard zero outside the sphere;  $active \Leftrightarrow r = \text{effective\_t}(t)$ 
5:   swap  $lattice\_curr \leftrightarrow lattice\_draft$ 
6:   POLARIZATIONTICK  $\triangleright$  elected-momentum broadcast over the  $W$ -ledger; stamps arrival ticks  $b(x)$ 
7:   for  $w = 0 \dots W-1$  do  $\triangleright$  loop over the internal dimension
8:     for each cell  $(x, y, z)$  in the 3-torus do
9:        $C \leftarrow lattice\_curr(x, y, z, w)$ ;  $D \leftarrow C$ ;  $M \leftarrow lattice\_partner(x, y, z, w)$   $\triangleright M$ : the partner
   cell
10:      if  $k < ENCOUNTER$  then
11:        ENCOUNTER( $C, D, M$ )  $\triangleright$  interaction rules R1–R6, Algorithms 2–3; source-kind
   (K/S/D/P) contact re-emission and momentum impulse
12:      else if  $k < GSLOT\_Z$  then
13:        glider slots  $\triangleright$  reserved for the antipodal transport schedule; not exercised by the
   reference simulation algorithm
14:      else if  $k < DIFFUSION$  then
15:        DIFFUSE( $C, D$ )  $\triangleright$  orphan expansion ( $a \leftarrow W$  from a full neighbor); homing  $homB \rightarrow$ 
   translation vector  $\mathbf{c}$ ;  $\mathbf{c}$  and  $kB$  transport; affinity settle
16:      else if  $k < RELOC$  then
17:        RELOCATE( $C, D$ )  $\triangleright$  consume the translation vector  $\mathbf{c}$  along the three axes (north,
   west, down)
18:      else if  $k < REISSUE$  then
19:        REISSUE( $C, D$ )  $\triangleright$  reset  $kB, homB, bB$ ; propagate affinity outward from  $active$  cells;
   consume the contraction flag  $cB$  (reset  $t$ )
20:      else
21:        FLOOD( $C, D$ )  $\triangleright$  synchronize the light clock:  $D.t \leftarrow \min$  of the six neighbor  $t$ 
22:      end if
23:       $D.k \leftarrow (C.k + 1) \bmod FRAME$ 
24:      if  $D.k = 0$  then  $\triangleright$  a new light frame begins
25:         $D.t \leftarrow (C.t + 1) \bmod (2 RMAX)$   $\triangleright$  island cells ( $a=W, t \leq RMAX$ ) count without the
   modulus
26:      end if
27:    end for
28:  end for
29:  APPLYCHARGECONJUGATION  $\triangleright$  matter–antimatter turnaround hook; exact no-op when the bias
 $\varepsilon = 0$ 
30:  COMMITSOURCETICK  $\triangleright$  commit source changes; on the frame boundary resolve unique internal
   contacts
31:  APPLYMOMENTUM  $\triangleright$  consume free photon pairs at turnaround; translate layer states by
   pending steps, preserving bound-pair identity and  $\mathbf{m}$ 
32:  SWAPLATTICES  $\triangleright$  draft  $\rightarrow$  curr; at  $k = 0$  refresh the partner ( $M.f \leftarrow \text{effective\_t}(M.t)$ ); while
 $k < ENCOUNTER$  rotate the partner  $W$ -slices (ROTATEPARTNERS)
33: end procedure

```

Algorithm 2 Exact interaction rules for the six charge bits (superposing bubbles).

```

1: function CANFORMPAIR( $a, b$ )
2:    $ca \leftarrow a.ch; cb \leftarrow b.ch$ 
3:   if  $ca = 000000B$  and  $cb = 000000B$  then                                ▷ R3: neutrino
4:     return true
5:   else if  $ca = 111111B$  and  $cb = 111111B$  then                                ▷ R4: antineutrino
6:     return true
7:   else if  $\overline{ca} = cb$  then                                                    ▷ R1: graviton, all six bits complementary
8:     return true
9:   end if
10:   $(q_a, w_{1,a}, w_{0,a}) \leftarrow \text{decode bits of } ca; (q_b, w_{1,b}, w_{0,b}) \leftarrow \text{decode bits of } cb$ 
11:   $col_a \leftarrow ca \& 111B; col_b \leftarrow cb \& 111B$ 
12:  if  $q_a \neq q_b$  and  $w_{1,a} = w_{1,b}$  and  $w_{0,a} \neq w_{0,b}$  and  $col_a \oplus col_b = 111B$  then    ▷ R2: gluon
13:    return true
14:  else if  $q_a = q_b = 0$  and  $w_{1,a} = w_{1,b} = 0$  and  $w_{0,a} = w_{0,b} = 1$  and  $col_a = col_b$  and  $col_a \notin$ 
     $\{000, 111\}B$  then                                                            ▷ R5: up quark
15:    return true
16:  else if  $q_a = q_b = 1$  and  $w_{1,a} = w_{1,b} = 1$  and  $w_{0,a} = w_{0,b} = 0$  and  $col_a = col_b$  and  $col_a \notin$ 
     $\{000, 111\}B$  then                                                            ▷ R6: up quark
17:    return true
18:  end if
19:  return false
20: end function

```

external capture rates, the charge-sign dependence, or a Coulomb distance law: the mechanism remains a candidate, not a measured effect.

7.1.3 Static magnetic interaction

The magnetic case uses the same photonic-pair mediation as Coulomb, but gates on sB instead of pB , and applies to charged leptons, quarks and W bosons. Because drive pairs emphasize pB over sB , fast particles tend to acquire a component perpendicular to their direction of motion.

7.1.4 Neutrino interactions

Let $q_{0,1}$ be a complementary electric-bit pair. A neutrino fragment has charge content $q_{0,1} = 00000B$; an antineutrino has $q_{0,1} = 01111B$. (Anti)neutrino fragments interact only with (right-)left-handed particles.

7.1.5 The up quark and down quark fragments

The color combinations of fragment pairs are shown in Table 3. (Anti)*quark* fragments have non-trivial color and are marked as $1q$ or $1\bar{q}$, (anti)neutrinos are ν and $\bar{\nu}$, gluons are g , and photons are γ . Since half the gluon fragments are right-handed, they are shown as photons. The shaded cells (same-color pairs) with both quarks carrying electric charge $q = +$ may form an *up quark* fragment. An *up quark* u can be any of the $+R+R$, $+G+G$ or $+B+B$ pairs, while a *down quark* d can be any of the $-R$, $-G$ or $-B$ single fragments. A proton fragment uud has an uncompensated color, balanced dynamically by the matter and antimatter ends of gluons. Hence the *electron* fragment must be $-N, -N, -N$.⁴ The number of up-quark fragments in each sector sets an upper limit on proton, and hence *atom*, production.

7.2 Self-interference

Space in this model retains a memory of particle trajectories, producing self-interference reminiscent of the double-slit experiment [11]. The trace is encoded by the last affinity value a and the most recent wave phase f left in visited cells. During encounter (Sect. 6.4), incoming and residual bubbles are compared. If $d = t$, standard cohesion occurs; if $d = f$, only the new bubble is reissued, altering the fermion's dynamics and creating interference patterns. This temporally shifted interaction among the constituent "bits" of qubits provides a substrate for complex quantum behavior in a discrete framework.

⁴This explains fractional quark charges.

Algorithm 3 cluster formation, electric/magnetic channels, and collapse (superposing and distinct bubbles).

```

1: function CANFORMCLUSTER( $a, b$ )
2:    $ca \leftarrow a.ch; cb \leftarrow b.ch$ 
3:   if  $ca = cb = 000000B$  or  $ca = cb = 111111B$  then                                 $\triangleright$  R3/R4 never cluster
4:     return false
5:   else if  $\overline{ca} = cb$  then                                                             $\triangleright$  R1 graviton never clusters
6:     return false
7:   end if
8:   decode  $(q, w_1, w_0)$  and  $col$  of  $ca, cb$  as in Algorithm 2
9:   if  $q_a = q_b$  and  $w_{1,a} = w_{1,b}$  and  $w_{0,a} = w_{0,b}$  and  $col_a = col_b$  and  $col_a \notin \{000, 111\}B$  then     $\triangleright$ 
    R5/R6 up quark never clusters
10:    return false
11:    else if  $a.reloc \neq 0$  or  $b.reloc \neq 0$  then                                     $\triangleright$  drive pair never clusters
12:      return false
13:    end if
14:    return  $w_{1,a} = w_{1,b}$  and  $q_a \neq q_b$  and  $w_{0,a} \neq w_{0,b}$  and  $col_a \oplus col_b = 111B$   $\triangleright$  only the R2 geometry
15:  end function
16: procedure ENCOUNTER( $C, D, M$ )
17:    $samePos \leftarrow C.x = M.x; sameT \leftarrow C.t = M.t$ 
18:   if  $samePos$  and  $sameT$  and CANFORMPAIR( $C, M$ ) then
19:      $D.kind \leftarrow M.kind \leftarrow P; D.pair\_idx \leftarrow M.w; M.pair\_idx \leftarrow C.w$      $\triangleright$  form a pair
20:      $D.a \leftarrow M.a \leftarrow$  shared affinity; REEMITATCONTACT( $D, M$ ); reset  $t$ 
21:     if  $newCount > 1$  and CANFORMCLUSTER( $C, M$ ) and  $C.pB = M.pB$  and  $C.sB = M.sB$ 
    and  $C.bB = M.bB$  then
22:        $D.bB \leftarrow M.bB \leftarrow \text{true}$                                              $\triangleright$  cluster
23:     end if
24:     else if  $C.pB$  or  $M.pB$  or  $C.sB$  or  $M.sB$  then                                 $\triangleright$  electric/magnetic contact
25:       if  $C.pB \wedge M.pB$  or  $C.sB \wedge M.sB$  then                                     $\triangleright$  collapse
26:          $D.kB \leftarrow D.cB \leftarrow \text{true}$                                         $\triangleright$  triggers reissue
27:       else
28:         ADIABATICEXCHANGE( $C, D, M$ )  $\triangleright$  exchange affinity and phase, relocate to partner center
29:       end if
30:     end if
31:  end procedure

```

	N 000	R 001	G 010	B 011	B 100	G 101	R 110	N 111
N 000	ν	$1q$ 1ℓ	$1q$ 1ℓ	$1\bar{q}$ 1ℓ	$1q$ 1ℓ	$1\bar{q}$ 1ℓ	$1\bar{q}$ 1ℓ	γ
R 001	1ℓ $1q$	$2q$	$2q$	γ	$2q$	γ	γ	$1\bar{\ell}$ $1q$
G 010	1ℓ $1q$	$2q$	$2q$	g	$2q$	γ	γ	$1\bar{\ell}$ $1q$
$\bar{B}$ 011	1ℓ $1\bar{q}$	g	g	$2\bar{q}$	g	$2\bar{q}$	$2\bar{q}$	$1\bar{\ell}$ $1\bar{q}$
B 100	1ℓ $1q$	$2q$	$2q$	g	$2q$	γ	γ	$1\bar{\ell}$ $1q$
$\bar{G}$ 101	1ℓ $1\bar{q}$	g	g	$2\bar{q}$	γ	$2\bar{q}$	$2\bar{q}$	$1\bar{\ell}$ $1\bar{q}$
$\bar{R}$ 110	1ℓ $1\bar{q}$	g	g	$2\bar{q}$	γ	$2\bar{q}$	$2\bar{q}$	$1\bar{\ell}$ $1\bar{q}$
$\bar{N}$ 111	γ	$1q$ $1\bar{\ell}$	$1q$ $1\bar{\ell}$	$1\bar{q}$ $1\bar{\ell}$	$1q$ $1\bar{\ell}$	$1\bar{q}$ $1\bar{\ell}$	$1\bar{q}$ $1\bar{\ell}$	$\bar{\nu}$

Table 3: Color combination (rows $\times$ columns = the two fragment colors). Legend: $1q$ = one quark, 1ℓ = one lepton, $2q$ = a quark pair (up quark), ν = neutrino, γ = photon and g = gluon; a bar denotes the antiparticle ($1\bar{q}$ = antiquark, $1\bar{\ell}$ = antilepton, $\bar{\nu}$ = antineutrino). Since half the gluon fragments are right handed, they are shown as photons. The shaded cells mark the same-color pairs that, with electric charge $q = +$, may form an *up quark* fragment in Orbis (cyan, matter colors) and its anti-up (yellow, antimatter colors).

8 Emergent island formation and the scaling of W

This section expands Sect. 4.1 and details how W -islands organize dynamically, and whether that dynamics selects a preferred population at all.

8.1 Objects and identity

A $3D$ island is a spatially localized collective of equal-charge constituents identified by its chief K . Each delegate D refers to that chief through its parent identity. Affinity does not identify an island: multiple islands and their dressing may share one affinity, and affinity is not an eligibility condition for chief election. A complete *particle* is a 3D island of bare charge plus a surrounding *pair dressing*:

$$\text{particle} = 3D \text{ island} + \text{pair dressing}. \quad (7)$$

The internal non-spatial dimension is partitioned into N_I identical W -islands, each containing

$$\ell = \frac{W}{N_I} \quad (8)$$

positions. A W -island is not an object in physical space; it is an internal charge family. All ℓ copies inside one island share the same six charge bits $(q, w_1, w_0, c_2, c_1, c_0)$, but each copy carries a distinct pB/sB pattern. This built-in multiplicity is why identical particles are so similar, while the distinct polarizations allow the radial Hofer pattern to emerge. Initially each internal address carries a singleton (S). Equal-charge singletons from different seed families may elect the same chief through encounter; neither their family index nor their affinity restricts this eligibility. Subsequent capture can therefore mix immutable W addresses from different families within one spatial island. The proposed role of photon-mediated information transport is to enable further island formation and mixing over time; this process remains to be demonstrated numerically. A formed island has the form

$$1K + nD. \quad (9)$$

In an admissible $D \times D$ encounter between equal-charge delegates, one delegate is promoted to a new chief K . Equality of charge is the only pair-specific eligibility condition: promotion does not require equal affinity, a common parent, a particular spin, or additional spatial separation beyond the general encounter conditions. The promoted source becomes its own chief, while the other source remains a delegate. This transition permits new island identities to arise after the initial singleton election; its presence must not be replaced by unconditional propagation of an existing chief identity.

Conversely, an admissible equal-charge $K \times K$ clash leaves one chief and demotes the other to a delegate of the surviving chief. In the reference experiment, the smaller W address survives; among several encountered chiefs, the smallest eligible address is retained. This is an explicit deterministic tie-break, not a demonstrated physical origin of symmetry breaking. Existing delegates of a demoted chief are not reassigned remotely by this transition. The propagation of their updated membership and the resulting balance between chief creation and removal remain subjects of the coagulation experiment.

Here n counts delegates associated with the chief; the drive pair dressing is excluded. Whether the aggregation dynamics selects a *preferred* population $1 + n$ —and whether such a preference is the same for every family, or varies with the lattice—is the open question this section tests. Nothing of the kind follows from complete membership of a seed family, and no value is imposed by initialization or election: the seed declares how many addresses *share* a family (ℓ), never how many of them should end up inside one island.

The chief gives the island its identity. Each bubble retains its intrinsic address w ; its parent and auxiliary leader identity record its association with a chief. Sharing affinity alone neither joins two islands nor makes an S a delegate. Counting chiefs and their delegates measures associations; spatial localization and persistence must be measured separately before those associations can be called stable islands. The seed bookkeeping identity is

$$W = 3L^2 = N_I \ell, \quad (10)$$

which supplies no independent bound of N_I on the number of dynamically formed chiefs. N_I is fixed by the partition of the ledger, the chiefs are produced by the encounter dynamics, and the reference runs show the two settling far apart rather than tied to each other: the number of chiefs is a measured outcome, not a multiple of the declared families.

8.2 Is the island population quantized?

A 3D island is an open dynamical object: it continuously captures free S bubbles and loses surface bubbles. Let $\Gamma_{\text{cap}}(N)$ and $\Gamma_{\text{esc}}(N)$ be the effective capture and escape rates for an island of characteristic population N . The size-selection hypothesis was originally stated as the condition that

$$\Gamma_{\text{cap}}(N) > \Gamma_{\text{esc}}(N) \quad \text{for small } N, \quad (11)$$

$$\Gamma_{\text{cap}}(N) < \Gamma_{\text{esc}}(N) \quad \text{for large } N. \quad (12)$$

A stable attractor N_* then appears near

$$\Gamma_{\text{cap}}(N_*) \simeq \Gamma_{\text{esc}}(N_*), \quad (13)$$

without any global counter. The characteristic charge of a particle would be the statistically preferred population of this attractor, dressed by the required photon pairs. The coordinated translation of Sect. 6.5 is what prevents premature deterioration of such localized particles across scales, from a resting aggregate up to parton-level probing. Because a kick displaces a single bubble while the mean center of mass advances only by $\Delta_{\text{CoM}} = n/N$, no constituent ever outruns the affinity that recaptures it. Identity survives every collapse-and-reissue because the intrinsic address w is immutable while only the auxiliary `leader_w` converges. Robustness is thus distributed over homeostasis (N_*), continuous affinity re-propagation at reissue, and conservative encounter rules — not stored in any fragile global state.

Status: measured negative for a size-selecting attractor; the only population quantum found is 2 (reference) or 3 (candidate), rule-induced, not geometric. The rate-crossing formulation is *degenerate* in this rule set: at the frozen plateau both capture and escape rates are *zero* (no delegates leave a frozen island), so $\Gamma_{\text{cap}}(N_*) = \Gamma_{\text{esc}}(N_*) = 0$ selects N_* by construction rather than by a dynamical balance. The only quantization this rule set produces therefore comes from the *fixed-point structure of the membership transitions*, derived from the reference rules themselves with no partition parameter involved (`experiments/DYNAMIC_QUANTIZATION_DERIVATION.md`).

1. The membership transitions T1–T5 read only the numerical order of w , the charge word ch , the parent linkage, and the source kind; none of them reads ℓ , `ISLAND_SIZE` or W . The partition enters only through the seed address map ($\lfloor w/\ell \rfloor$ giving the family index, `initSim.cpp`) and through the two candidate macros, which is why their outcome is a declared multiplicity rather than a derived quantum.
2. A frozen co-located state has at most one non-singleton group per charge word, and every such group has population ≤ 2 . The bound follows from T2 (delegate promotion): any group of population ≥ 3 contains two delegates of the same charge word, which meet within W frames and promote one to chief, reducing group size by one. The maximum is reached, not merely satisfied: $K = W - 8$, $D = 8$ at both $L = 9$ ($235 + 8 = 243$) and $L = 15$ ($667 + 8 = 675$).
3. At $L = 9$, the canonical superposed-seed reference census (closed gate, $S = 16384$) freezes at $K = 235$, $D = 8$, $S = 0$, with maximum group population 2 and no group at any larger population; the open-gate ($S = 64$) census gives 145 $K + 5$ D plus 93 pair halves, also with maximum population 2 (archived log `experiments/census_ref_e19_s64/census.csv`: frames 2–4 identical at $K = 145$, $D = 5$, $P = 93$). Both are reproduced exactly by the reference build.
4. At $L = 15$ ($W = 675$, $RMAX = 7$), the same reference dynamics yields $K = 667$, $D = 8$, with the population quantum again 2 and no group above it, confirming the bound at a second lattice size with no candidate macro involved.

A constructed balance rule that switches T2 off *inside* one family (the candidate macro `DD_INTRA_ISLAND_FIX`) changes only the population from 2 to 3, because the rule still uses T1/T3/T4/T5 across families. No rule in the repository produces a larger quantum without *reading* the seed’s multiplicity explicitly in its transition logic, which makes such a result axiomatic rather than emergent. The geometric-quantum hypothesis (shell, ball, and contact capacities) is separately tested and falsified in `experiments/GEOMETRIC_QUANTUM.md`: no stable population matches any of the shell capacities $\{26, 66, 158, \dots\}$, the ball capacities $\{27, 93, 251, \dots\}$, or the contact capacity W ; the realized population histogram over all 59 group-census files is a near-continuum from 1 to 33 with no quantization gaps. The last positive path is closed: *at $L \leq 15$ the model has no population quantum at values other than the rule-induced 2, 3, or 5.*

Flux-harness confirmation. The independent statistical instrument in `quantization/` tests the same attractor hypothesis from per-event fluxes $\Gamma_{\text{cap}}(N)$ and $\Gamma_{\text{esc}}(N)$ via OLS regression and permutation-surrogate testing (`quantize_stdlib.py`). On the synthetic mean-reverting control the harness recovers $N^* = 30.38$, 95% CI [29.87, 30.91], surrogate $p = 0.005$ (PASS). On the six real datasets from the repository *all four frozen configurations return zero escapes and are reported INCONCLUSIVE*: the reference at $L = 9$ and $L = 15$, the `EXCLUSION+DD_INTRA_ISLAND_FIX` candidate, and the `ADDRESS_TARGET_FSM` candidate. Where turnover exists (the directional-channel candidates), the fitted slope is $b = +0.494$ – $+0.495$, i.e. *anti-restorative*—larger islands grow faster, so the population is driven upward rather than bounded. The flux harness thus confirms the membership-theorem verdict from an independent statistical angle: no population attractor exists in any real run in the repository at the accessible sizes.

Circulation $\mathbf{J}$ as a membership invariant (measured). After the population-attractor line closed, a separate program asked whether a conserved circulation on the island could do work that a pile of members cannot. Define the integer spin of a group by

$$\mathbf{J} = \sum_{\text{members}} \mathbf{r} \times \mathbf{m},$$

with $\mathbf{r}$ the toroidal offset of a member from its chief and $\mathbf{m}$ that member’s momentum vector. The definition reads only w , parent linkage, source centers and $\mathbf{m}$; it contains no L , no partition parameter and no `ISLAND.SIZE`. Three pre-registered steps have been run (`experiments/J1.SPIN.md`):

- **J0 (baseline).** Across every archived census in the repository (~ 3200 groups), $\mathbf{J} = \mathbf{0}$ identically. The cause is structural: in the reference path body elements keep $\mathbf{m} = \mathbf{0}$, so no circulation is generated.
- **J1 (protection).** A candidate gate (`SPIN_GATED_FSM`) refuses delegate escape while the chief’s $\mathbf{J} \neq \mathbf{0}$. In a frozen-rotor tube where the only agent is the surface-escape rule, the $J = 0$ arm releases both delegates and the $J \neq 0$ arm holds the group ($1K + 2D$, zero escapes), matching the four registered predictions exactly. Circulation is inert until a rule consults it; once consulted, it decides membership.
- **J2 (selection).** The same circulation protects an already planted size; it does not produce a $dN-N$ rate crossing or select N_* . Steps that would test conservation of $\mathbf{J}$, scaling with cavity radius, and mixed charge words remain open.

Thus $\mathbf{J}$ is the first measured quantity in the project that can hold a group together without reading the seed’s partition, but it is a *protection* invariant, not a population quantum. It does not restore a size-selecting quantum; at most it can help keep an island that some other mechanism has already assembled.

8.3 Origin of W and FSM addressing

The non-spatial dimension has size $W = 3L^2$, whose quadratic scaling is motivated in Sect. 3, and its ledger factorises as $W = N_I \ell$ into N_I charge families of $\ell = W/N_I$ copies each (Sect. 4.1). In a finite-state machine implementation the W coordinate is therefore not a single integer accessed by modulo; it is a pair of registers, an island index in $[0, N_I)$ and a copy index in $[0, \ell)$. This preserves the model’s FSM character: all addressing and state transitions remain local and involve only comparison, copying, and small counters. Both integers are evaluated once at initialization from the declared topology, and none of the aggregation results below depends on their particular value.

8.4 Candidate mechanisms for the encounter’s transport

The reference build leaves the transport channel closed, and the trace of Sect. 6.7 on small lattices measures why. At frame 1 the W sources are one singleton each at the lattice center, sharing a single clock value, so the ledger is decided by the first transition: by frame 2 it has already reached its plateau ($K = W - 8$, $D = 8$ at $L = 7$) and it does not change again within the era, while the shell follows the exact torus counts (1, 26, 66, 158 cells per layer for radii 0 to 4). The polarization pair is identically zero and no axis is ever elected—(pol_u, pol_v) = $\mathbf{0}$ and $\mathbf{m} = \mathbf{0}$ for all W layers in every frame measured, at both sieve settings—because the zero-polarization seed is the fixed point of the election, broadcast and reconstruction loop of Sect. 6.4: no axis can be elected without an existing pair, and none can be

reconstructed without arrival stamps—and not a single layer is displaced in the seven eras of a control run read at $L = 7$.

What does displace a bubble is the encounter’s pair walk: the two partners each step one cell toward the other, so the step is the sign of the shortest offset to the partner, and the partner is taken from the W ladder. Measured on the era-1 cascade at $L = 7$, in the transport-enabled build where the polar seed is elected (so that $\mathbf{m} \neq \mathbf{0}$ and pairs are actually reconstructed there): 94 layer displacements, of which *none* follows the layer’s own color octant and 79 differ from it in exactly one sign—with the channel closed, as in the reference build, the same cascade displaces none of them. The pair channel transfers the partner’s direction across octants.

Four candidate mechanisms address this. All are compiled off in the reference build, each behind its own macro, and measured on lattices of side $L \leq 9$:

- **CHARGE_DISPERSION_FSM.** While no axis is elected, step each layer along the octant of its own charge word ($c_2c_1c_0$ read as the three coordinate signs), and stop as soon as $\mathbf{m} \neq \mathbf{0}$. Measured: the W layers resolve into eight blocks, Orbis ($w_1 = 0$) at negative y and Umbra at positive y , the center of mass at the population imbalance, and 147 of 147 displacement signs matching the octant.
- **POLAR_SEED_FROM_PLACEMENT.** Closes the fixed point by electing the source center itself when the tournament finds no candidate. Measured: $\mathbf{m} \neq \mathbf{0}$ on every layer, with 13 671 cells carrying a reconstructed pair.
- **PAIR_SAME_OCTANT** admits the walk only between partners that share the color triplet, and **PAIR_OWN_AXIS_EXCHANGE** keeps the reaction’s magnitude but gives it the direction of the receiver’s own octant. Measured together over twelve eras of a single run at $L = 7$: the ordering the dispersal established survives, the gap between the two sector means falling from 3.1 cells at era 1 (−1.5 and +1.6) to 1.4 at era 4 (−0.9 and +0.5) and then *flooring* at a few tenths of a cell instead of vanishing (−0.4 and −0.0 at era 5, −0.1 and +0.2 at era 12). It fails once: over a three-frame window of era 11 the Orbis mean sits at the lattice center and the Umbra mean below it (+0.0 and −0.2), and the next era’s cascade repairs it. The Umbra mean crosses the lattice center at era 5 and returns, so what is conserved is the ordering, not each sector’s side of the center, and the center of mass wanders up to half a cell, so the state is not stationary—whereas in the same build without these two rules the gap has collapsed to zero by era 3 and oscillates around it, inverting at era 4. At $L = 9$ the same comparison is decided inside the first era: the build without the two rules ends era 1 already inverted (+0.1 and −0.4) while the pair-rule build holds the ordering with a gap of 1.4 cells. The ledger stays at its plateau throughout, and the flight displacements are octant-aligned at every cascade measured (147 of 147 at the dispersal, 32 of 32 at the era-1 cascade); the flight steps of later eras that fall away from the octant are impulses inherited from earlier frames rather than a second transport path, and the single-axis island cohesion steps are counted separately.

These are candidates rather than results of the reference build: none of the macros is enabled there, the measurements above come from lattices of side $L \leq 9$, and nothing is claimed here about the transport at the lattice size the appendix takes for the fabric.

At this point, the core construction of the cellular automaton is complete.

9 Conclusion

9.1 Undecidability and long-term dynamics

Long-term dynamics involve Poincaré cycles. Because the CA can be Turing-equivalent, questions about recurrence versus chaos are undecidable in general [12, 13]. The finite state space guarantees eventual recurrence (Pigeonhole principle), but the configuration space grows exponentially with L , making detection infeasible [14, 15]. Affinity further stretches the probable recurrence time: by binding bubbles into persistent particles and aligning them into long-lived 3D islands, it reduces the effective number of independently evolving configurations and channels the dynamics through organized attractors rather than allowing rapid, unstructured mixing. Black-hole fusion may then absorb exceptional configurations (‘Gardens of Eden’) and drive the system toward a Poincaré cycle. Undecidability limits algorithmic predictability, so analytical tools are needed for general insights.

9.2 Falsifiability of the physical model

A direct observation that an electron physically traverses both paths in a double-slit experiment would contradict the model. Self-interference here arises from recorded lattice traces, not from simultaneous path occupation.

9.3 Novelty of the work

Compared with prior work surveyed in Sect. 2, this model introduces four main novelties:

1. A non-spatial dimension W that acts as a dynamic computational layer, enabling cross-layer encounter—a superluminal but strictly non-signaling and non-manipulable update mechanism that protects causality—and richer interactions than Zuse’s purely spatial “digital particles.”
2. Six charge bits per cell, giving Lie-group-like symmetries and Noether-like conservation at a programmable level.
3. Affinity, which groups bubbles into particles and encodes self-interference histories.
4. Ontological collapse events that break strict reversibility within a Poincaré cycle, matching the effective irreversibility of quantum measurement.

The model retains a discrete lattice while capturing quantum and relativistic features; it remains exploratory and must still be validated against empirical data.

9.4 Limitations and open problems

For clarity, we collect here what the current implementation cannot yet do. These are structural limitations of the code as it stands (small lattice, reference-modulus electroweak gate), not of the model idea; each is stated with the corresponding evidence in the full manuscript.

- **No measured bound-state properties yet.** The in-simulation census registers no pair formation at the reference parameters ($\text{pairM} = \text{pairA} = 0$, $\text{form} = 0$, $\text{cluster} = 0$), and the controlled two-bubble experiment yields a quantitative null result there: of 6,996 overlapping cell-frame events only 18 (0.26%) pass the $s2B$ sieve and none triggers the pair branch. The parameter sweep demonstrates that the pair branch *can* fire ($\text{form} > 0$ for $S = 4096, 128, 64, 32$ at $L = 7$ and $S = 64$ at $L = 9$), so the pair-formation channel is now a demonstrated mechanism in a parameter regime; the interaction phenomena it would mediate at the reference settings, and the properties of the produced pairs (binding energies, decay rates, cross-sections), remain unmeasured. Bound states such as hadrons and atoms therefore remain conjectural; the aggregates that do persist (the $1K + nD$ islands of Sect. 8) are affinity-driven populations, not charge-bound composite particles.
- **Continuum limit absent.** The model is defined on a finite 3-torus of side $L \leq 31$ with a single cell as the unit of length. There is no procedure for taking $L \rightarrow \infty$ (or the cell size to zero), no coarse-graining scheme, and no derivation of a partial-differential or field-theory limit. The $r \cdot \sin(r)$ wave cloud (Sect. 5.2) is an emergent lattice pattern, not a solution of a continuum equation.
- **No finite-size scaling study.** Every measurement in this paper comes from $L \leq 31$ (in practice $L = 7, 9, 15$) and no asymptotic regime is analyzed. The rule carries no length scale to renormalize, so what is missing is not a renormalization but a finite-size scaling study of the dimensionless observables—the $1K + nD$ membership, the population quantum, the capture and escape rates, the plateau ratio K/W —at several lattice sizes. The only scaling relation measured so far is the saturation of the ledger in the frozen plateau, $K + D = W = 3L^2$, exact at $L = 9$ and $L = 15$; nothing here shows that the properties measured at those sizes survive at the lattice size that Appendix A takes for the fabric.
- **No quantitative comparison with measured cross-sections.** No scattering amplitude, cross-section, or decay rate has been computed from the automaton, and no quantitative particle-physics prediction follows from the model’s rules. Units are defined only up to the lattice scale X (Appendix A); there is no mapping from cells-per-tick to physical velocities or energies.

- **Small-lattice and boundary effects.** The reported runs use $L = 7$; the spherical cavity and the radial dead-zone at $r = \text{RMAX}$ suppress corner artifacts by construction, but also truncate long-range phenomena. The layer-symmetric seed makes all 75 islands statistically identical, a configuration choice rather than a property of the dynamics.
- **Electroweak channel dormant at the reference modulus only.** The $s2B$ sieve admits fewer than 1% of overlaps at the reference amplitudes and modulus, so weak phenomena—neutrino interactions, W/Z fragments, decay chains, and the inter-sector rules of Sect. 6.4.2—are not observed at the default settings; the sweep shows the channel firing at coarser moduli, so the reference dormancy is a parameter choice, not a structural impossibility.
- **Open problems.** (i) Measure the properties (binding energy, lifetime, scattering behavior) of the pairs produced by the open $s2B$ channel, and study how the pair-formation rate depends on S , U_0 , L and the impact parameter; (ii) define a continuum limit and derive effective equations; (iii) turn the charge-combination census into genuine cross-section predictions with a unit mapping; (iv) scale the lattice to test multi-particle dynamics beyond two bubbles; (v) derive the model constants (U_0 , sieve thresholds, `phase.full`) rather than setting them by hand.

These limitations delimit the scope of the present work: the manuscript reports mechanisms and exploratory observations, not a predictive physical theory. The falsifiability discussion above should be read in this light.

9.5 Future inclusions

Ongoing work simulates the geometric structures that emerge from these local, multi-layered updates. Preliminary tests already show that the collective dynamics of fragments produce a continuous $r \cdot \sin(r)$ density cloud, localized wave-packets, stable spiral patterns and macroscopic rotations directly from discrete bit distributions. These observations support the conjecture that the universal fabric can bridge discrete local rules and the smooth symmetries of physical spacetime. A further inclusion would be the headless trace harness behind the measurements of Sect. 8.4: it advances the model without the graphical application and prints one line per light frame (shell radius and cell count, the attractor census, the charge-word roster, the per-class displacements, and the alignment of each displacement with the layer’s own charge octant). It builds with `nmake trace-first-era` and takes the lattice side, the sieve modulus and the frame count on the command line, together with the candidate mechanisms of that subsection as compile-time options, so those numbers can be re-obtained independently of the reference configuration. The reference configuration and that candidate configuration are one build switch apart in the accompanying repository (`nmake` or `nmake CANDIDATES=1`), so either can be built from the same sources without editing a file.

9.6 Final thoughts

The model is an economical ontological framework⁵ operating at a deep subatomic scale. Observable reality is approximate and emergent, with continuous spacetime symmetries (Lie, Lorentz, diffeomorphism, CPT, etc.) arising from the dynamics. Each particle is composed of an enormous number of bubbles—each a miniature universe—so intrinsic non-locality provides a natural substrate for superposition-like behavior and qubit analogs. The theory remains to be refined computationally and tested against experiment; whether it can become a predictive, falsifiable, *bona fide* physical theory is an open question.

⁵Especially when contrasted with the Many-Worlds Interpretation of quantum theory [16].

Nomenclature

Symbol	Meaning	Units
L	lattice side (integer; the physical ledger takes it a multiple of 64)	cells
W	internal dimension size, $W = 3L^2$	layers
X	cell size	m
k	housekeeping (global) tick	dimensionless
t	wavefront (local) tick	dimensionless
f	wave phase, $f = \text{effective_t}(t)$	dimensionless
N_t	ticks per light frame	ticks
$v_{\max}$	maximum information speed, $v_{\max} = X/N_t$	cells per tick
c	emergent speed of light (observer reading of $v_{\max}$)	m s^{-1}
r, r^2	dynamic radius and squared radius from the source	cells, cells ²
u, v	wave displacement and quadrature	dimensionless
$\text{pol}_u, \text{pol}_v$	reconstructed transverse polarization pair	dimensionless
pB, sB	polarization sign bits	boolean
$s2B$	electroweak sieve test bit	boolean
$\text{hom}B, \text{c}B, \text{k}B, \text{b}B$	homing, contraction, collapse, cluster flags	boolean
$b(x)$	broadcasted arrival stamp of the elected momentum	ticks
ch	six-bit charge word ($w_1 w_0 q c_2 c_1 c_0$)	bits
a	affinity (not spatial island identity)	dimensionless
$\mathbf{m}$	momentum vector of a source ($ \mathbf{m} = L/2$ typical; immutable under inertia)	cells
reloc	consumable translation impulse (filled by encounter, applied by <code>applyMomentum</code>)	cells
C	source-center coordinate	cells
kind	source kind (K, S, D, P)	dimensionless
S	electroweak sieve modulus (<code>s2b_target</code>)	dimensionless
$P(u, S)$	sieve trigger probability	dimensionless
N^*	equilibrium W-island population	bubbles
$\Gamma_{\text{cap}}, \Gamma_{\text{esc}}$	island capture and escape rates	bubbles per frame
O	observable-universe diameter	m
Ed	Eddington number (proton count)	dimensionless
SEP	initial bubble separation in the scattering runs	cells

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During the preparation of this work, the authors used generative AI tools to improve the readability and language, as well as to assist in debugging and optimizing numerical algorithms. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- [1] J. A. Wheeler. “Information, physics, quantum: the search for links”. In: *Complexity, Entropy and the Physics of Information*. Ed. by W. Zurek. Reading, MA: Addison-Wesley, 1990.
- [2] S. Wolfram. *A New Kind of Science*. pp. 23–60, 112, 865–866. Wolfram Media, 2002.
- [3] K. Zuse. “Rechnender Raum”. In: *Elektronische Datenverarbeitung* 8 (1967), pp. 336–344.
- [4] E. Fredkin. “An Introduction to Digital Philosophy”. In: *International Journal of Theoretical Physics* 42.2 (2003), pp. 189–247.
- [5] N. Margolus. “Universal Cellular Automata Based on the Collisions of Soft Spheres”. In: *Collision-Based Computing*. Ed. by A. Adamatzky. London: Springer, 2002.
- [6] G. ’t Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*. Fundamental Theories of Physics. Series edited by H. van Beijeren et al. Berlin: Springer, 2016.
- [7] Stephen Wolfram. *The Wolfram Physics Project*. 2020. URL: <https://www.wolframphysics.org> (visited on 04/05/2025).
- [8] J. Hardy, Y. Pomeau, and O. de Pazzis. “Time Evolution of a Two-Dimensional Model System. I. Invariant States and Time Correlation Functions”. In: *Journal of Mathematical Physics* 14.12 (1973), pp. 1746–1759.
- [9] U. Frisch, B. Hasslacher, and Y. Pomeau. “Lattice-Gas Automata for the Navier–Stokes Equation”. In: *Physical Review Letters* 56.14 (1986), pp. 1505–1508.
- [10] A. Furtado Neto. *It from Bit – a Concrete Attempt*. 2024. URL: https://github.com/automaton3d/it_from_bit.
- [11] R. P. Feynman, R. B. Leighton, and M. Sands. *The Feynman Lectures on Physics, Volume 3: Quantum Mechanics*. See discussion on double-slit experiment in Chapter 1. Addison-Wesley, 1965.
- [12] Stephen Wolfram. *Theory and Applications of Cellular Automata*. Singapore: World Scientific Publishing, 1983.
- [13] P. Gács, L. Lovász, and M. Simonovits. “On the Computation of the Evolution of Cellular Automata”. In: *Acta Informatica* 11.2 (1979), pp. 149–159.
- [14] J. Bernstein. *The Pigeonhole Principle: A Problem Book in Mathematics*. Princeton, NJ: Princeton University Press, 2007.
- [15] T. Toffoli and N. Margolus. “Cellular Automata Machines: A New Environment for Modeling”. In: *Nature* 343.6259 (1980), pp. 87–93. DOI: 10.1038/343087a0.
- [16] Hugh Everett. ““Relative State” Formulation of Quantum Mechanics”. In: *Reviews of Modern Physics* 29.3 (1957), pp. 454–462. DOI: 10.1103/RevModPhys.29.454. URL: <https://doi.org/10.1103/RevModPhys.29.454>.
- [17] P. Halpern and N. Tomasello. “Size of the Observable Universe”. In: *Advances in Astrophysics* 1.3 (Nov. 2016).
- [18] The ARCHEOPS Collaboration. “The Cosmic Microwave Background Spectrum from the Full Focal Plane Observation of ARCHEOPS”. In: *Astronomy and Astrophysics* 399 (2003), pp. L19–L23.

Appendix

A The scale of the lattice

This appendix fixes the two dimensionful parameters of the fabric—the cell scale X , that is the distance between neighbouring cells, and the lattice side L (Sect. 3)—from observation. Two cosmic relations are written down and solved together under two competing identifications of what a bubble occupies; one solution is refuted by the same observations that fix the other, and the surviving one determines X and L . Note that the calculations below are derivation-only: per the purity constraints (Sect. 3.3), real numbers serve solely as intermediaries to compute integer parameters and are never invoked inside the runtime transition rule. The values of X and L obtained here are rounded to integers and used once at initialization as fixed topological data. The empirical inputs used below—the measured speed of light c , the observed diameter of the universe, the proton rest mass and the baryon count—are observer-measured quantities: c is the emergent speed read from the fabric’s fundamental constant $v_{\max} = X/N_t$. The transition rule is scale-free: it fixes L , N_t and $W = 3L^2$ as dimensionless numbers, and the cell scale X does not appear in it, so the physical size of a cell has to be imported from observation. The calibration is one-way: the observed quantities enter, and the two free parameters X and L are solved from them; nothing in this appendix is derived by the transition rule. Two independent cosmic relations are used. The first is geometric: the radius of the spherical cavity in which the rule evolves its bubbles is identified with the radius of the observable universe, Eq. (14)—an identification of scale. The second is material: the baryon and photon content of the universe must fit into the internal addresses that the rule provides, Eq. (15)—an identification of capacity. Neither follows from the transition rule and neither is tested by it; together, once the internal bookkeeping is declared, they fix the pair (X, L) . The subsections below keep the levels apart: A.1 and A.2 state the two identifications, A.3 works out their consequences, A.4 applies the one observational test this appendix has, and A.5 collects the inputs and the assumptions. The radius of the observable universe [17], half of $O = 8.8 \times 10^{26}$ m, is compared with the radius of the cavity that the rule evolves, $L/2 = \text{RMAX}$ cells at scale X (Sect. 5.2):

$$\frac{O}{2} = \frac{L}{2} X, \quad \text{that is} \quad O = L X. \quad (14)$$

Only the radius is identified: it is the extent the spherical shell actually reaches, whereas the diagonal of the enclosing cube is not a length the rule ever uses.

A.1 The radius relation

The active shell expands from the center to the amplitude $L/2 = \text{RMAX}$ and contracts back, one cell per light frame (Sect. 5.2), so the cavity radius is $LX/2$ and Eq. (14) equates it with $O/2$. The relation ties the two unknowns together without fixing either: it gives the product

$$L X = O,$$

so that a larger cell implies a smaller lattice side, and vice versa. Nothing in the rule fixes this identification; the alternatives open to it—the side, the circumference or the diagonal of the enclosing cube—change X by the square root of the chosen length, that is by a factor below two, so the order of magnitude obtained below does not depend on which one is taken.

A.2 The content relation

The second relation counts the baryon and photon content of the universe and asks how many places the rule offers for it. The average CMB photon frequency [18] is

$$[f] = 160 \text{ GHz} = 1.6 \times 10^{11} \text{ Hz},$$

which, converted to energy, gives

$$E_{\text{CMB}} = h[f] = 1.060171224 \times 10^{-22} \text{ J}.$$

The proton rest energy in Joules is

$$m_p c^2 = 1.503277592969 \times 10^{-10} \text{ J},$$

so the number of average CMB photons energetically equivalent to a proton is

$$n_p = \frac{m_p c^2}{E_{\text{CMB}}} = 1.4179573 \times 10^{12}.$$

With the Eddington number $Ed = 10^{80}$ as the proton count, the total proton contribution in photon units is

$$n_{pT} = 4 Ed n_p = 5.67183 \times 10^{92},$$

to which the $n_\gamma = 4 \times 10^{84}$ photons of the universe add nothing at this precision:

$$n_{\gamma T} = n_\gamma + n_{pT} = 5.67183 \times 10^{92}.$$

The factor 4. Ed counts one family of carriers—the protons—and the census that the lattice has to hold is four times that number. The charge word makes two independent binary distinctions (Sect. 3): the weak bit w_1 selects the internal sector, Orbis ($w_1 = 0$) or Umbra ($w_1 = 1$), and the remaining bits select either a fragment or its complement, so matter and antimatter arise in pairs by construction rather than through a built-in asymmetry. A baryon census in Orbis therefore implies the mirrored antibaryon census, and the same census again in Umbra, which coexists spatially with Orbis (Sect. 6.1); the pair rules treat the two sectors on the same footing, the two up-quark rules differing only in the value of w_1 (R5 for $w_1 = 0$, R6 for $w_1 = 1$; Sect. 6.4). The four families—matter and antimatter, in Orbis and in Umbra—are what $4Ed$ counts. This is a modeling assumption rather than a measurement: it asserts that the lattice carries the four families in step with one another.

A photon of frequency $[f]$ spans $n_b = [f]/f_0$ bubbles, where $f_0 = c/(LX)$ is the frequency of one bubble; with the factor 2 for the pair of bubbles that constitutes a photon (Sect. 6.1), the total bubble content is

$$n_{bT} = n_b n_{\gamma T} = 2 \frac{[f] n_{\gamma T} LX}{c} = 2 \frac{[f] n_{\gamma T} O}{c},$$

the last form using Eq. (14). What remains to be fixed is where those bubbles go. The places the rule offers are the internal addresses, $W = 3L^2$ (Sect. 3), and the ledger factorises as $W = N_I \ell$ into N_I charge fragments of ℓ addresses each (Sect. 8). In the fragment count used here,

$$N_I = 192 L, \quad \ell = \frac{L}{64}, \quad N_I \ell = 192 L \times \frac{L}{64} = 3L^2 = W :$$

the number of charge fragments is $192 L$ and each is replicated $L/64$ times, so the fragments fill the internal addresses exactly; with L taken a multiple of 64 those copies are integers, so the ledger partitions the W addresses rather than stating a ratio. Identifying one bubble with one internal address gives the second relation,

$$n_{bT} = N_I \ell = 3L^2 = \frac{2[f] n_{\gamma T} O}{c}, \tag{15}$$

in which the CMB frequency $[f]$ cancels, so that only the baryon content, the proton mass, c and the observed radius survive. The relation no longer involves X : it fixes the lattice side from the content alone.

What a bubble is, in this counting. Three things are assumed, and they are worth stating because the counting is otherwise ambiguous. First, n_{bT} is an *instantaneous inventory*: bubbles are the places occupied at one light frame, each counted once, with no account of how long a bubble lives, of several bubbles at one site, of reuse, or of internal state; a rate would be a different quantity. Second, the time scale of a bubble is the era, not the tick: the shell expands from the center to $\text{RMAX} = L/2$ and back in L light frames, each advancing one cell, so one era lasts LX/c and the fundamental frequency is $f_0 = c/(LX)$ —the f_0 used above, so no separate frequency scale is introduced. Third, dividing the baryon budget by the number of bubbles fixes an *energy per bubble*, $h f_0 = hc/(LX)$, one quantum of the cavity's fundamental mode; in the rule a photon's frequency is an integer (the number of superposed pairs, Sect. 6.2), and reading that integer in hertz is precisely the step that f_0 supplies. Each of the three belongs to the calibration, not to the rule.

A.3 The lattice side

The address-based reading of Eq. (15) gives the lattice side directly,

$$L = \left(\frac{2[f] n_{\gamma T} O}{3c} \right)^{1/2} \approx 1.3 \times 10^{61}, \quad X = \frac{O}{L} \approx 6.6 \times 10^{-35} \text{ m},$$

together with $W = 3L^2 \approx 5.3 \times 10^{122}$ internal addresses, $L^3 \approx 2.4 \times 10^{183}$ spatial cells, and $N_I = 192L \approx 2.6 \times 10^{63}$ charge fragments carrying $\ell = L/64 \approx 2.1 \times 10^{59}$ addresses each. Both values are order-of-magnitude illustrations of what Eqs. (14) and (15) imply—identifications rather than derived relations—and the sensitivities of Sect. A.5 move them by factors of order unity. Two counts are available for the places a bubble may occupy—the internal addresses W and the spatial cells L^3 —and they differ by a factor L ; the subsection below tests the choice against observation.

A.4 What the observations exclude

This is the only step of the appendix in which observation does work; everything above is algebra on identifications. A lattice of cell size X can represent only wavelengths longer than twice the spacing: the discrete wave update of Sect. 5 fills a Brillouin zone whose shortest wavelength is $2X$, so the highest mode energy is of order $\hbar c/2X$ and no structure finer than a cell can exist in the fabric. Read as a constraint, this says that the lattice must be fine enough to carry the shortest wavelength observed, $X \leq \hbar c/E_{\max}$. The highest-energy particles observed, $E_{\max} \sim 10^{20}$ eV, give

$$X \leq \frac{\hbar c}{E_{\max}} \approx 2.0 \times 10^{-27} \text{ m}, \quad (16)$$

and the protons of the LHC (6.5 TeV) already give $X \leq 3.0 \times 10^{-20}$ m.

The natural alternative fails this test by a wide margin. If the places a bubble may occupy are the spatial cells rather than the internal addresses—that is, $n_{bT} = L^3$, which ignores the internal partition—Eq. (15) gives instead $L \approx 8.1 \times 10^{40}$ and a cell of $X = O/L \approx 1.1 \times 10^{-14}$ m ≈ 11 fm, and that cell has its zone edge at $2X \approx 22$ fm, that is $\hbar c/2X \approx 9$ MeV, so the lattice could carry neither a pion ($\lambda = 1.46$ fm, a seventh of a cell) nor a 6.5 TeV proton ($\lambda = 3.0 \times 10^{-5}$ fm, 3.6×10^5 times shorter than a cell); a cosmic ray of 10^{20} eV has $X/\lambda \approx 5.5 \times 10^{12}$, so the dispersion parameter $(X/\lambda)^2$ reaches 3.0×10^{25} exactly where the observed absence of energy-dependent arrival times constrains it far below unity; and Eq. (16) is exceeded by a factor 5.5×10^{12} —twelve orders of magnitude. Counting spatial cells is therefore not marginally wrong but refuted, and with it the picture in which the available places are the cells of the lattice.

The reading used above passes the same test with room to spare. Its cell is 6.6×10^{-35} m, which is 3.3×10^{-8} of the wavelength of a 10^{20} eV cosmic ray, so $(X/\lambda)^2 \approx 1.1 \times 10^{-15}$: the dispersion lies far below anything those observations can detect, every known particle is resolved (the zone edge sits at $\hbar c/2X \approx 1.5 \times 10^{21}$ MeV), and the cavity radius $LX/2$ reproduces the observed $O/2$ by construction. The two readings use the same observed content and the same radius relation and differ only in what counts as a place for a bubble; nothing in the relations singles out the internal addresses, so the discriminating step is the exclusion above: what it leaves standing is the reading in which the places are the internal addresses— $N_I = 192L$ charge fragments of $L/64$ addresses each, multiplying to the declared $W = 3L^2$ —rather than the spatial cells.

The argument assumes what the model asserts: that the fabric's waves are the physical fields and that X is their granularity. Under that reading counting spatial cells is refuted and the internal addresses stand. If instead the lattice were a computational device with no physical cell, no calibration could be attempted in the first place, since the meter would never enter the rule.

A.5 Inputs, status, and sensitivity

The two relations rest on inputs of two kinds, and the comparison above turns on the distinction. Measured or observed quantities:

H1 $O = 8.8 \times 10^{26}$ m: the diameter of the observable universe [17], whose half is identified with the cavity radius in Eq. (14).

H2 c : the emergent speed, read from the fabric's fundamental constant $v_{\max} = X/N_t$.

H3 m_p : the proton rest mass (CODATA).

H4 $Ed = 10^{80}$: the Eddington number of protons in the universe, an order-of-magnitude baryon census.

H5 $[f] = 160$ GHz: the average CMB photon frequency [18]; it cancels in Eq. (15) and is quoted only for the intermediate chain.

Modeling assumptions, none of them an output of the rule:

A1 the carrier census is four times the baryon census—matter and antimatter, in Orbis and in Umbra (the factor 4 above);

A2 the address ledger $W = N_I \ell = 3L^2$ read as $N_I = 192 L$ charge fragments replicated $\ell = L/64$ times each (Sect. 8);

A3 one bubble per internal address, $n_{bT} = N_I \ell = 3L^2$, in Eq. (15). The alternative to A3—one bubble per spatial cell—is the reading refuted above.

Sensitivity: the reading used above gives $\delta \ln L = \frac{1}{2}(\delta \ln O + \delta \ln Ed + \delta \ln m_p)$ and $\delta \ln X = \frac{1}{2}\delta \ln O - \frac{1}{2}(\delta \ln Ed + \delta \ln m_p)$, so a factor-2 uncertainty in any single input moves both L and X by $2^{1/2} \approx 1.4$. The cell-based reading moves L by $2^{1/3} \approx 1.3$ and X by $2^{2/3} \approx 1.6$, which is why its failure cannot be blamed on the inputs: the factor 5.5×10^{12} by which it misses Eq. (16) would take some forty doublings of a single observed quantity.

Scale. The rule is dimensionless, so it defines the dynamics for every L ; every *measured* statement in this paper, however, comes from $L \leq 31$, and no asymptotic analysis connects that regime to the value obtained above. The only measured scaling relation available is the saturation of the ledger in the frozen plateau, $K + D = W = 3L^2$, exact at $L = 9$ ($235 + 8 = 243$) and at $L = 15$ ($667 + 8 = 675$); since $D = 8$ in both, that plateau is made of the seed's singletons and is not evidence of aggregation at large L . Extrapolated to $L \approx 1.3 \times 10^{61}$ it would give $W \approx 5.3 \times 10^{122}$ places—about 10^{42} bubbles per baryon family of the census used above, which is consistent with reading a particle as an enormous number of bubbles but far beyond anything the simulations can exhibit—and were the plateau relation to hold at that size, the number of islands would be of the same order as the number of addresses, which cannot be reconciled with reading a particle as one island. Whether the properties measured at $L \leq 31$ (the $1K + nD$ membership, the population quantum of 2, the windows that scale with W) survive at that size is therefore an open problem: this appendix reports the identifications and their consequences, not a scaling law.